Supporting Information for

Energetic Prioritization and Neurometabolic Scaling in a Social Insect Brain

Zach N. Coto[1*], Dajia Ye[2], Sara Arganda[3,4], Jon F. Harrison[5], James F.A. Traniello[1]

[1]Department of Biology, Boston University, 5 Cummington Mall, Boston, MA 02215, USA

[2]Department of Biology, Stanford University, Stanford, CA 94305, USA

[3]Departamento de Biología, Universidad Rey Juan Carlos, Móstoles 28933, Spain

[4]Instituto de Investigación en Cambio Global (IICG-URJC), Universidad Rey Juan Carlos, Tulipán s/n, 28933 Móstoles, Spain

[5]School of Life Sciences, Arizona State University, Tempe, AZ 85287-4501, USA

* Zach N Coto

Email: zcoto@bu.edu

This PDF file includes:

Supporting materials and methods

Supplementary Figures 1 to 17

Supplementary Table 1 to 15

Supporting materials and methods

Supplementary methods

Colony Husbandry

Laboratory colonies were maintained in conditions promoting colony health with minimal disturbance. Lab colonies were housed in artificial nests made from 30 x 25 x 10 cm Fluon®-lined plastic boxes and were provided with test tubes (22 x 147 cm) half-filled with water and fitted with tight cotton plugs. Colonies were maintained on a 12L: 12D cycle at 55% humidity and 25.0˚C

and fed *ad libitum* carbohydrate (1M sucrose solution) and sources of protein and fats (mealworms, fruit flies, scrambled eggs) several times weekly. Workers from lab colonies were sampled when the colonies were approximately three years old and had several thousand workers.

Measurement of Brain Metabolism

We use a standard concentration of glucose and pyruvate in the media of brains measured across all treatments to reduce the likelihood that substrate availability rather than physiological condition would affect brain metabolic rate. Preliminary studies demonstrated that brain metabolic rate did not increase under increasing media glucose concentrations (5mM, 10mM, 50mM, 100mM), suggesting that brain metabolic rates are not limited by the glucose concentration used *ex vivo*. In the only study to have measured hemolymph carbohydrate concentrations in starved ant workers (1), workers deprived of carbohydrates for 11-13 days averaged 5 mg ml^{-1} (28mM), nearly three times higher than our media glucose concentration, suggesting that starved brains are not limited by glucose concentrations *in vivo*.

Seahorse XF HS mini cartridges were prepared according to Agilent protocols. Oxygen consumption rate (OCR) was recorded at 31±0.03˚C, consistent with the thermal preference of *T. immigrans* (2). For each brain, OCR was recorded for five 3-minute periods, with 1 minute of mixing time between each measurement. The average OCR of the five recordings was used for analysis. Our analysis excluded brains with unstable metabolic rates to avoid inclusion of brains that may have been cryptically damaged during excision. We calculated the standard deviation of five recordings for each brain, the interquartile ranges (IQR) of the standard deviations, and defined outliers as those with a standard deviation in brain OCR exceeding 150% of the IQR. Ten

outliers (5 food-deprived, 3 controls and 2 queenright of a sample size of 166) were excluded from analyses of worker mass, metabolic rates and movement.

Inclusion of the outliers did not affect statistical significances (Table S15); body metabolic rates were still depressed more by starvation than brain metabolic rates. When outliers are included, mean brain mass-specific metabolic rate of food-deprived workers (16.6 ± 4.4 μW) is decreased by approximately 24% from control values (21.7 ± 7.8 μW), while body mass-specific metabolic rate of food-deprived workers (2.3 ±1.1 μW) is decreased by 44% from controls (4.1 ± 1.5 μW).

Brain metabolic rates were recorded within 2.5 hours following excision. Brain OCR remained stable for minimally 2.5 hours, indicating robust viability *ex vivo*. This may result from the ability of the insect brain to store significant levels of substrates (3) available to alleviate energy limitation during food deprivation. Representative traces of brain OCR are provided in Figure S12. We did not measure worker respiratory quotients (RQ), which in other species have been reported as 0.91, 0.71, 0.77, 0.92, and 1.02, with an average value of 0.87 (4-8). We therefore converted brain OCR to joules assuming a RQ of 0.87, with 20.5 joules ml^{-1} oxygen (9), and 22.4 L mol^{-1} oxygen.

Previous studies support our use of *ex vivo* brain metabolic rates to estimate *in vivo* treatment effects. Buschiazzo et al. (10) recorded *in vivo* glucose uptake in rat brains using PET imaging of capillary glucose concentrations and *ex vivo* glucose uptake by suspending excised brains in glucose media and using autoradiography to measure glucose concentrations in tissue slices. *In vivo* and *ex vivo* brain glucose uptake was recorded from rats that had either been fed or starved for 48 hours. *In vivo* brain glucose consumption rates (controls: 119.5 nmol min^{-1} g^{-1}, starved: 46.1 nmol min^{-1} g^{-1}) were both approximately double *ex vivo* brain glucose consumption

rates (controls: 61.1 nmol min^{-1} g^{-1}, starved: 28.3 nmol min^{-1} g^{-1}). Additionally, Oikawa and Itazawa (11, 12) measured the metabolic rates of a large fraction of minced fish body tissues, including the brain *in vitro* and demonstrated that summing the *in vitro* tissue metabolic rates generated a metabolic rate one-third to one-half of a whole live fish's resting metabolic rate. This is expected, as all activity (heart rate, peristalsis, e.g.) is stopped in the minced *in vitro* tissues. Therefore, if our *ex vivo* tissue metabolic rates differ from *in vivo* metabolic rates as a result of eliminating synaptic inputs, previous studies on *ex vivo* tissue metabolism in vertebrates suggest that our values underestimate *in vivo* metabolic rates. If this is the case in ants, the fraction of whole-body metabolic rate due to brain metabolism is underestimated in our study, suggesting our primary conclusion that brain metabolic rate is prioritized under energetic stress is conservative and supported by our study.

Measurement of Body Metabolism.

Whole-body CO_2 emission rates were measured over 60-90 minutes, similar to the durations over which brain metabolic rates were measured *ex vivo*. We note that these are not resting metabolic rates, but include periods of movement and positional stability, and thus reflect effects of nutritional stress on behavior. Workers were weighed to within 10 μg using a Mettler AE 163 analytical balance and placed in respirometry chambers constructed from 2 ml syringes attached to three-way stop-cock valves and syringe needles. CO_2 - free air from a Linde zero-air gas cylinder was allowed to flow at 50 ml min^{-1}, regulated by an Omega mass flow regulator, through a respirometry chamber for 1-2 minutes, sufficient for all CO_2 produced by a worker to be flushed and establish a baseline % CO_2 near zero. The chamber was then sealed to a volume of one ml using the syringe plunger for approximately 60-90 minutes at 31±0.19°C. Next, a bolus of 0.8 ml

air from the chamber was injected through a gasket into an air stream leading to the gas analyzer at a flow rate of 50 ml/min.

A worker's bolus of CO_2 generated a peak of $\%CO_2$. We corrected the baseline of $\%CO_2$ to 0 and obtained the volume of the CO_2 ($volCO_2$) accumulated within the chamber containing the worker by integrating the peak $\%CO_2$ vs the x-axis (minutes) using Eq. **1**:

$$\mathrm{Vol}CO_2 = \int (\%CO_2 \times 0.01) \times (\text{Flow rate, ml min}^{-1}) \quad [\mathbf{1}]$$

We corrected for the fact that only 0.8 ml of the 1ml syringe volume was injected and for the impact of worker volume on the volume of CO_2 using Eq. **2**:

$$\text{Corrected } \mathrm{Vol}CO_2 = \mathrm{Vol}CO_2 \times (1.2) \times \left(\frac{\text{chamber vol.} - \text{worker vol.}}{\text{chamber vol.}}\right) \quad [\mathbf{2}]$$

where 1.2 accounts for the additional 0.2 ml not injected and worker body volume is calculated according to Lighton et al. (1987) as (1.01 × worker body mass in grams). $VolCO_2$ was divided by the duration, in minutes, that the worker was sealed within the chamber to calculate worker body CO_2 emission rate (ml min^{-1}). This was converted to metabolic rates assuming a respiratory quotient of 0.87 as described above for brains.

It is possible that respiratory quotients (RQ) could shift with food-deprivation, as found in vertebrates, though a previous study of the impact of seven days of starvation on worker metabolism found no effect on the RQ of anaesthetized ants (4). However, if RQ varied with treatment in our study, the most dramatic, physiologically-realistic scenario would be that the controls had an RQ of 1.0 (pure sugar metabolism) and food-deprived workers had an RQ of 0.7 (pure fat metabolism). Such a change would require an alteration in the joules per ml of oxygen from the 20.5 value we used for all workers to 21.2 for controls and 19.6 for the food-deprived workers. The metabolic rate of controls would then be 3.4% higher than our reported value and the metabolic rate of food-deprived workers would be 4.5% lower than our reported value. We

hypothesized (and observed) a reduction in body metabolic rate with food deprivation, assuming no change in RQ with nutritional status. Therefore, if food deprivation shifted metabolism toward fat rather than sugar use, the reduction in body metabolic rate would be larger than we calculated, and so our estimate of the energetic prioritization of the brain is conservative.

Syringes were nominally air-tight, but some minor CO_2 leakage from the atmosphere into the syringe was noted. We corrected for CO_2 leakage by recording 12-14 measurements of CO_2 volumes infiltrated into each empty respiratory chamber over a range of times after sealing. We calculated SMA regressions in R (13) for leaked CO_2 volume vs time, and then used the duration of the respirometry measure of the worker in that syringe to calculate the fraction of CO_2 measured for the worker that is attributable to leak and subtracted these from the Corrected $VolCO_2$ calculated using Eq. **2** above. Statistics for these regressions are presented in Table S14.

Because outliers in body mass-specific metabolic rate may result from leaks during measurement, we excluded 6 workers with outlier body mass-specific metabolic rates (2 fed controls, 4 queenright), which were defined and calculated in the same method as for calculating outliers in brain OCR variation. Statistical comparisons of our models including or excluding outliers are provided in Table S15. The inclusion of outliers did not affect the statistical significance of the any effect.

General Linear Mixed Model Diagnostics and Selection.

Different error distributions for a general linear mixed model were evaluated for each response variable prior to data analysis in order to generate models that met assumptions of either a Gaussian distribution (for normally distributed data) or an inverse-Gaussian distribution (for non-normal data) and which minimized the number of non-significant terms included in the model.

Terms included in our models were treatment (Queenright, Controls, and Food-deprived) and colony-collection environment (Field, Laboratory and Urban) as fixed effects and colony identity as a random effect. Additionally, because brain mass and body metabolic rate scaled with body mass (Figures S13-S14), we initially included body mass as a fixed effect for models of brain mass, brain metabolic rate, brain mass-specific metabolic rate, relative brain metabolic rate and body metabolic rate. Because brain metabolic rate scaled with body metabolic rate (Figure S15), we initially included body metabolic rate as a fixed effect for modeling brain metabolic rate and brain mass-specific metabolic rate. Brain metabolic rate did not scale with brain mass (Figure S16). However, because brain mass-specific metabolic rate exhibits a treatment-environment interaction effect, while brain metabolic rate does not, this suggested that brain mass may have an effect on brain metabolic rate and therefore brain mass was also included as a fixed effect in models of brain metabolic rate, brain mass-specific metabolic rate and relative brain metabolic rate.

For each response variable, we first generated a model including the above terms with a Gaussian distribution and determined whether the raw data fit Gaussian assumptions using a Kologorov-Smironov (K-S) test and dispersion test using the *DHARMa* package in R (14), a Levene's test to determine if data exhibited heteroskedasticity and a Shapiro-Wilks test to determine if residuals were normally distributed. If assumptions were satisfied, the model was simplified by sequentially dropping non-significant interacting terms followed by non-significant main effect terms to generate simpler models, validated using AIC estimates among possible models. When two terms were simultaneously insignificant, we dropped the term with the lowest partial eta^2 effect size, determined using the *effectsize* package (15) or *anova_stats* function in base-R. If two terms were both insignificant and had the same effect size, they were simultaneously dropped from the model. Terms were dropped until only significant terms remained. If non-

significant terms could not be dropped without the model violating assumptions of normality, the model was rerun with a non-normal, inverse-Gaussian distribution and the non-significant terms were again dropped sequentially as described above. If dropping non-significant terms again violated model assumptions, we applied either a square-root transformation, a log-transformation or, for proportions, a logit-transformation, re-ran the model with a Gaussian distribution and sequentially dropped non-significant terms as described above. If we were unable to remove all non-significant terms from any model, we selected the model with the lowest AIC score and proceeded with data analysis. Models tested for each response variable using the chosen error distribution are provided in Tables S5-S13. For reference, Tables S9 and S11 include presentation of several models with AIC scores lower than the chosen model. These alternative models were not chosen because they included non-significant effects that could be dropped from the model. However, use of these models would not change the interpretation of our results.

The above procedure was applied to all response variables other than behavioral and movement data. Behavioral response scores were non-parametric and rank-based using an ordinal scale. Therefore, we used a Kruskal-Wallis χ^2 test to assess impacts of treatment on worker behavioral responses. A Levene's test indicated equal variance across colony-collection environments ($F = 1.749$, $p = 0.177$), although there was unequal variance across treatment categories ($F = 4.54$, $p = 0.012$). Because many workers did not move during respirometry, worker movement data were zero-inflated and could not be modeled with a Gaussian or an inverse-Gaussian distribution. Therefore, a negative-binomial distribution accommodating zero-inflation was used to model worker movement with the *glmmTMB* package in R (16). For movement data, we also simplified the model by sequentially dropping non-significant terms.

Supplementary results

Worker Survival Probability Under Conditions of Food Deprivation

A Weibull AFT regression performed using treatment and environment as interacting effects indicated significant effects of treatment and colony-collection environment on worker survival probability, demonstrating that food deprived workers were significantly physiologically stressed relative to controls and that field-collected workers tended to have lower survival probability than workers collected from other environments. Controls exhibited significantly higher survival probability than food-deprived workers (hazard = -1.58±0.06, $Z = 26.12$, $p < 0.001$). Among controls, workers from field colonies exhibited lower survival probability than those from lab (hazard = -0.41±0.17, $Z = 2.36$, $p = 0.018$) and urban (hazard = -0.96±0.30, $Z = 3.19$, $p = 0.001$) colonies. Workers from lab colonies, however, did not differ in survival probability from workers from urban colonies (hazard = 0.54±0.28, $Z = 1.94$, $p = 0.052$). Among food-deprived workers, workers from field colonies exhibited lower survival probability than workers from lab colonies (hazard = -0.43±0.04, $Z = 11.95$, $p < 0.001$) and workers from urban colonies (hazard = -0.22±0.03, $Z = -6.53$, $p < 0.001$), and workers from urban colonies also exhibited lower survival probability than workers from lab colonies (hazard = -0.21±0.03, $Z = 7.12$, $p < 0.001$). During food-deprivation, workers from lab colonies survived longer than workers from other colony-collection environments, showing 80% mortality at up to 39 days following the start of treatment, compared to 17-20 days for workers from other colony-collection environments. This may have been due to workers from lab colonies having access to consistent *ad lib* feeding prior to studies. Despite variation in survival between colony-collection environments, these data nonetheless indicate that food-deprived workers experienced significant physiologically stress when sampled. Data are presented in Figure S11.

Metabolic Scaling Patterns

The slope of log brain mass on log body mass was significant for queenright workers (Figure S13, slope = 1.07, $r^2 = 0.203$, $p < 0.01$) and the slope of log body metabolic rate on log body mass was significant for queenright workers (Figure S14, slope = 1.37, $r^2 = 0.087$, $p = 0.02$) and food-deprived workers (Figure S14, slope = 1.44, $r^2 = 0.157$, $p < 0.01$). Food-deprived workers also exhibited a significant negative grade shift in body metabolic scaling relative to queenright workers (Figure S14, $W = 32.42$, $p < 0.01$). Brain metabolic rate scaled with body metabolic rate in queenright workers (Figure S15, slope = 0.73, $p = 0.04$), but did not scale with brain mass in any condition (Figure S16).

Colony Effects on Worker Body Mass, Brain Mass and Brain Metabolism

Controls exhibited significantly higher body mass than queenright workers and food-deprived workers. Because worker brain mass did not differ between treatments, the significant increase in body mass resulted in a significant decrease in relative brain mass in controls. Nutrition is normally regulated by both individual and colony energy requirements (17). The absence of colony regulation of feeding behavior by controls may have increased worker energy intake relative to queenright workers, resulting in increased body mass of controls. Alternatively, *Tetramorium* workers exhibit morphological variation independent of body mass (worker head width ranged from 0.72-1.08 mm), and this size variation could be hypothesized to contribute to significant differences in worker body mass across treatments. Head width scales with body mass (slope = 0.30 ± 0.02, $r^2 = 0.70$, $p < 0.01$). However, a linear mixed model using square root-transformed head width with treatment and colony-collection environment as fixed effects

indicates that only controls from lab colonies exhibit larger head widths than queenright workers ($T = 3.30$, $p < 0.01$) and food-deprived workers ($T = 3.32$, $p < 0.01$). These differences are not significant in either field workers (Controls vs Queenright, $T = 1.65$, $p = 0.23$; Controls vs Food-deprived, $T = 0.99$, $p = 0.59$) or urban workers (Controls vs Queenright, $T = 1.77$, $p = 0.18$; Controls vs Food-deprived, $T = 0.21$, $p = 0.98$), despite body mass varying consistently between treatments across all three environments. These results suggest that higher body mass of controls is only partially due to worker morphological variation and may be driven by the absence of social regulation of feeding behavior that is present in queenright colonies.

Significant treatment-environment interaction effects were only found for brain metabolic rate and brain mass-specific metabolic rate (Figures S5 and S6, Tables S2 and S3). Notably, there were differences between queenright workers and food-deprived workers from field colonies, as well as differences between queenright workers and controls from field colonies. However, there were no differences between controls and food-deprived workers and neither response variable exhibited a significant treatment or environment effect after accounting for the interaction effect (Table 1). Therefore, this demonstrates that although subcolony culture may affect brain metabolism of workers from specific colony-collection environments, experimental treatments had no significant effect on brain metabolism, despite impacting body metabolism and worker mortality.

Supplementary Figures

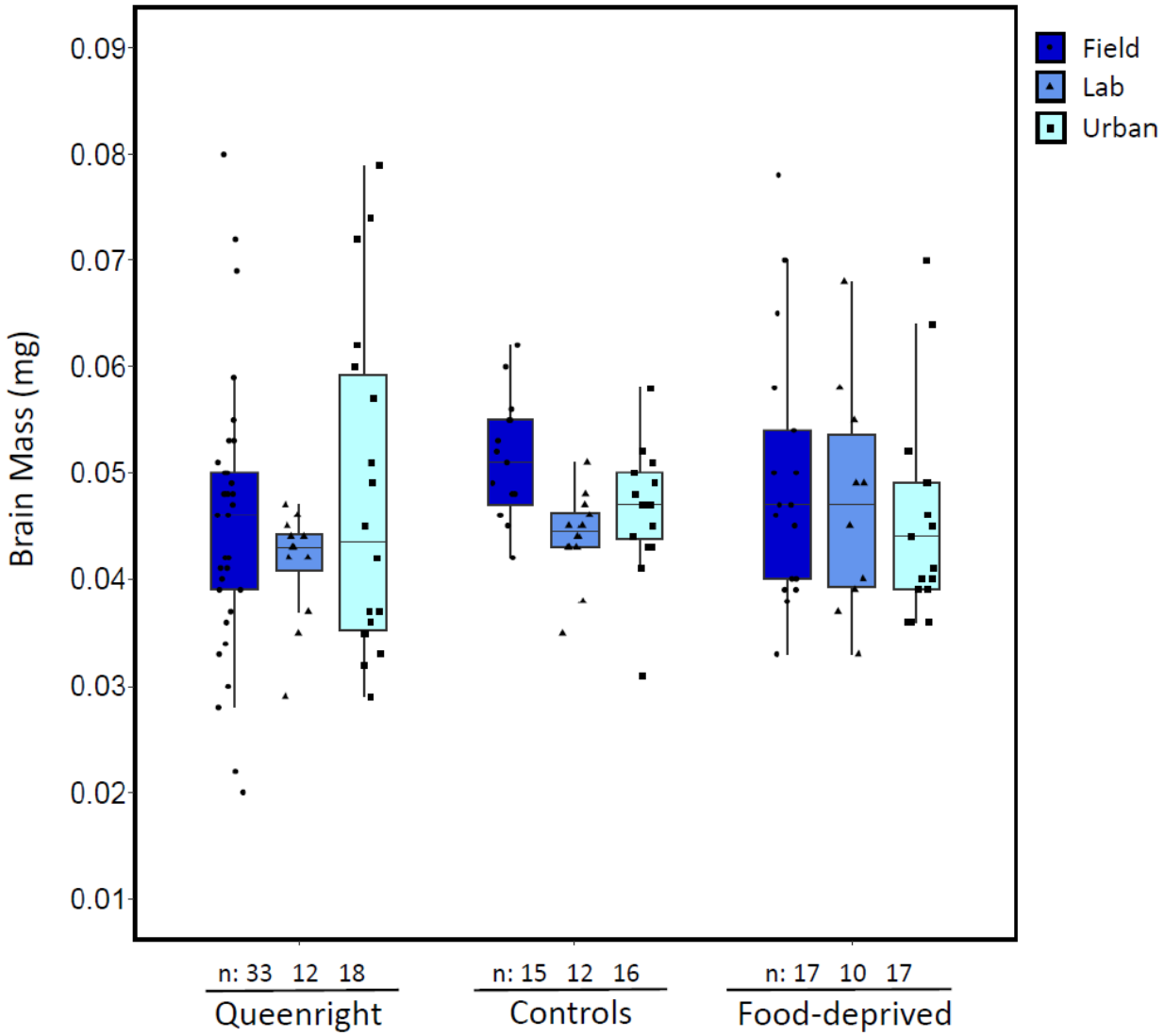


Fig S1. Colony-collection environment did not have a significant effect on brain mass (Table S5, $\chi^2 = 1.50$, $p = 0.47$) and there was no significant interaction with Treatment (Table S5, $\chi^2 = 4.72$, $p = 0.32$). n indicates sample sizes for Field, Lab and Urban colony-collection environments, respectively. Box plots represent medians ± 150% of the interquartile range

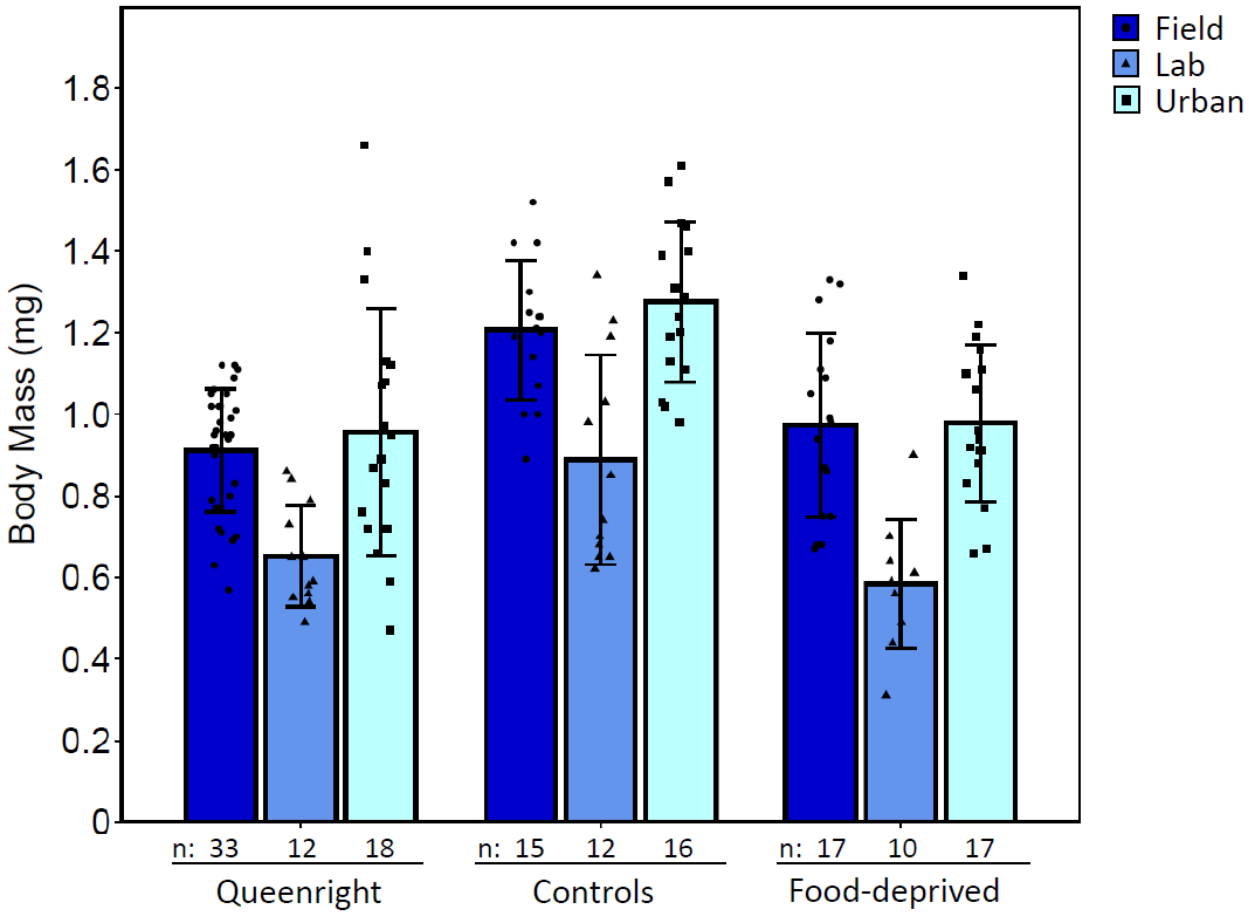


Fig S2. Colony-collection environment did not have a significant impact on worker body mass (Table S6, F = 2.14, p = 0.16) and there was no significant interaction with Treatment (Table S6, F = 0.46, p = 0.77). n indicates sample sizes for Field, Lab and Urban environments, respectively. Bar plots represent means ± standard deviation.

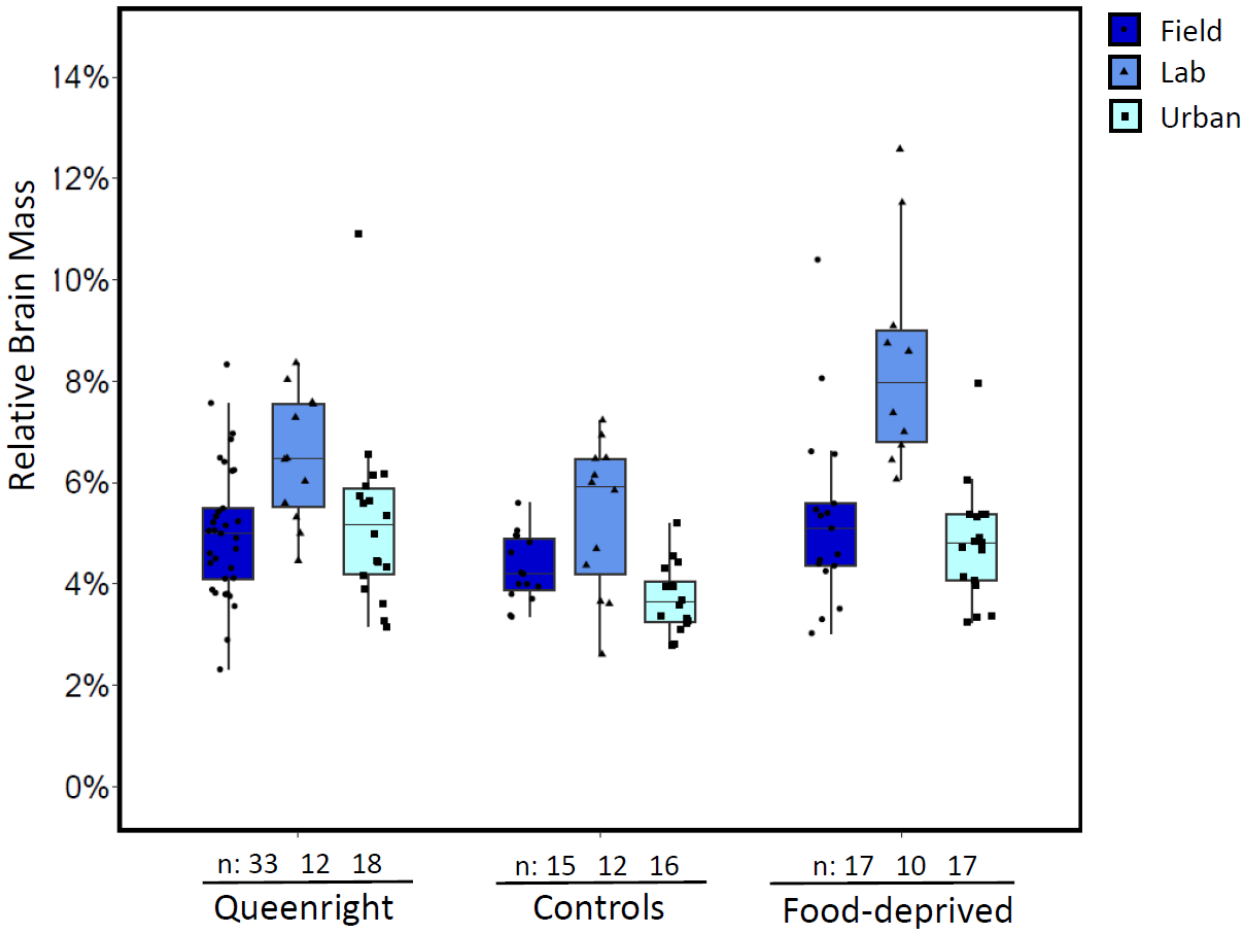


Fig S3. Colony-collection environment had a significant effect on relative brain mass (Table S7, $\chi^2 = 30.63$, $p < 0.01$). Workers from laboratory colonies had higher relative brain mass than workers from field colonies ($Z = 4.66$, $p < 0.01$) and urban colonies ($Z = 5.53$, $p < 0.01$). Workers from field and urban colonies did not differ ($Z = 1.44$, $p = 0.32$). There was no significant interaction of colony-collection environment with treatment (Table S7, $\chi^2 = 7.33$, $p = 0.12$). n indicates sample sizes for Field, Lab and Urban colony-collection environments, respectively. Box plots represent medians ± 150% of the interquartile range.

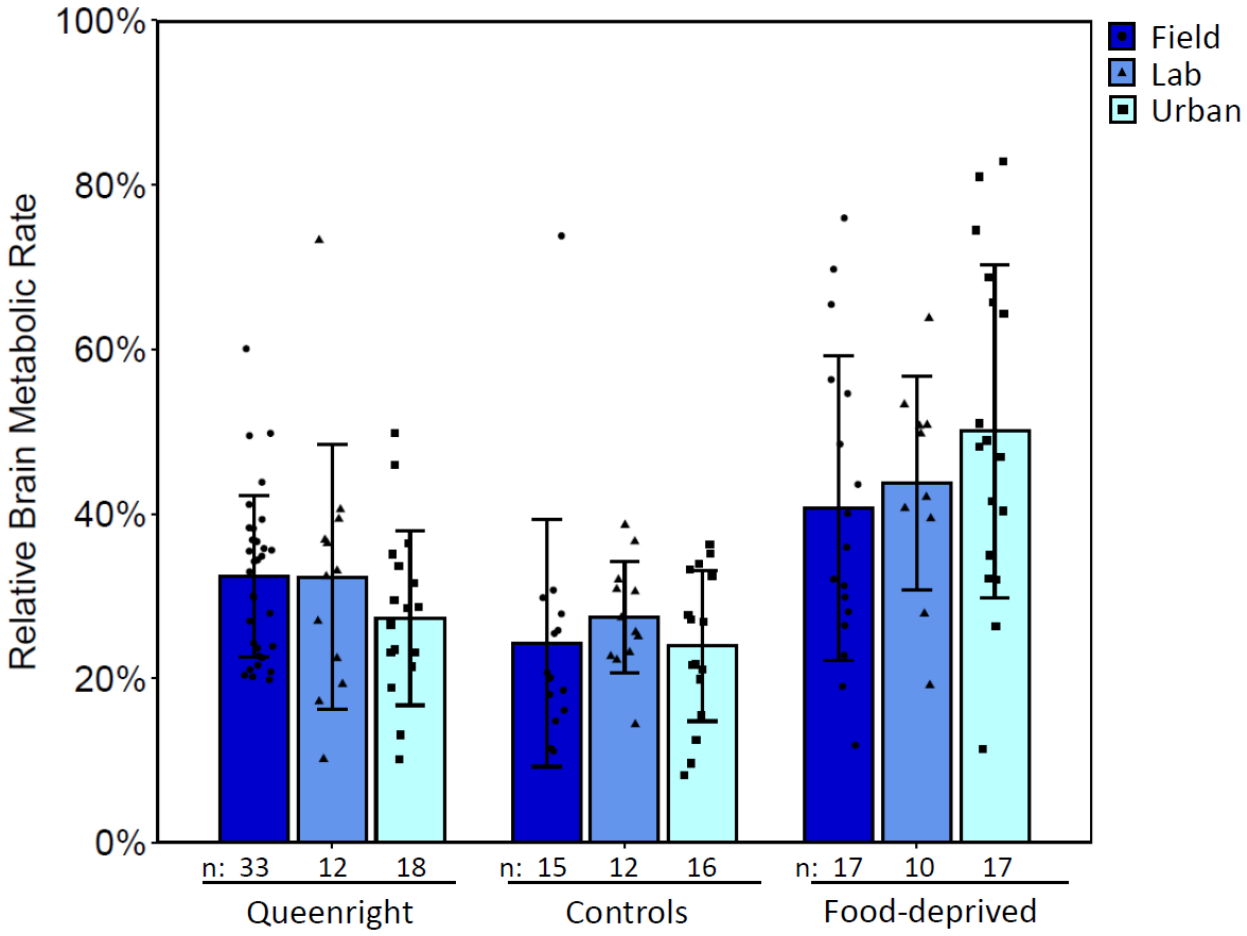


Fig S4. Colony-collection environment did not have a significant effect on relative brain metabolic rate (Table S8, F = 0.20, p = 0.82) and there was no significant interaction effect between treatment and environment (Table S8, F = 1.70, p = 0.15). n indicates sample sizes for Field, Lab and Urban colony-collection environments, respectively. Bar plots represent means ± standard deviation.

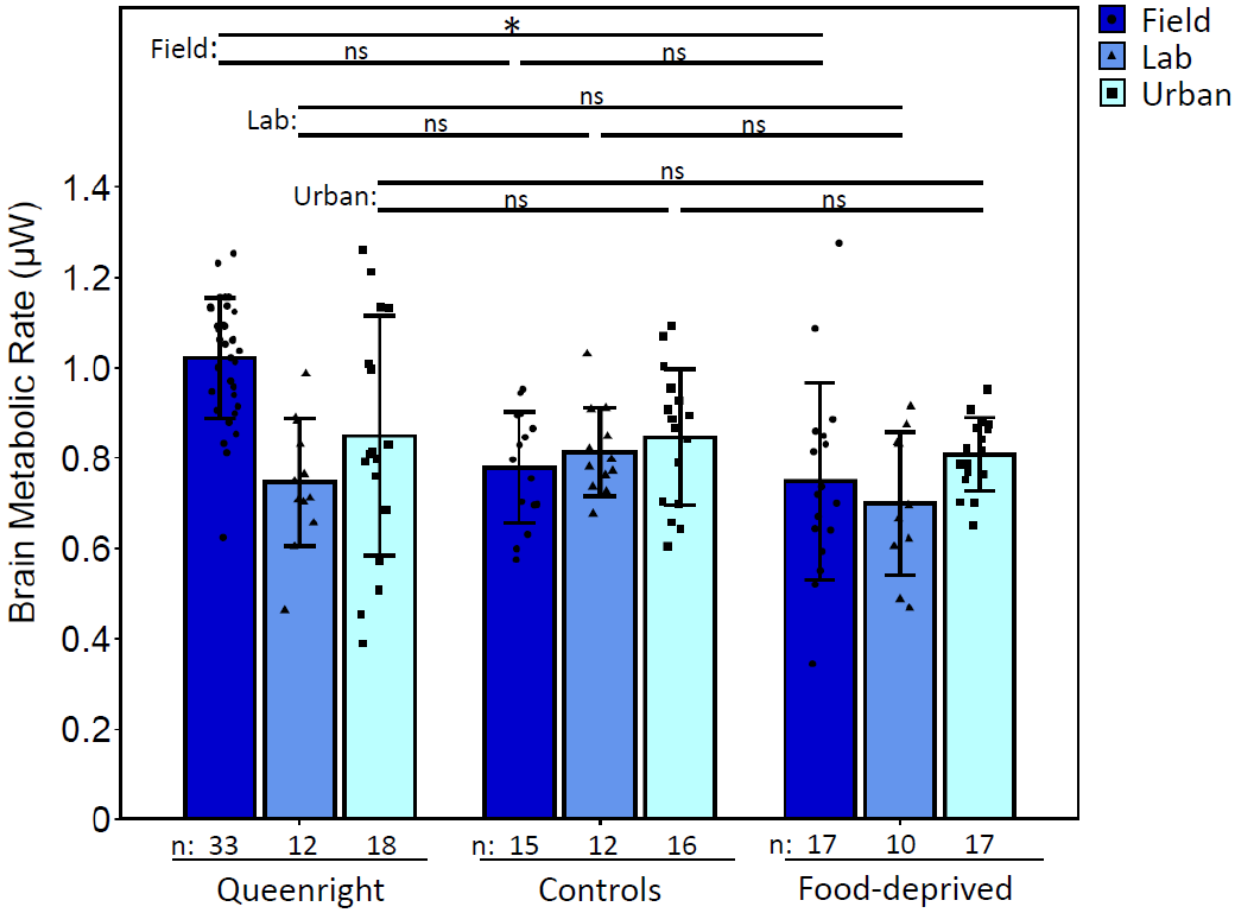


Fig S5. Colony-collection environment did not have a significant effect on brain metabolic rate (Table S9, $F = 0.26$, $p = 0.77$), but there was a significant Treatment-Environment interaction effect (Table S9, $F = 3.42$, $p = 0.01$). Among workers collected from field colonies, queenright workers exhibited higher brain metabolic rates than food-deprived workers ($T = 3.32$, $p < 0.01$). Statistics of all pairwise comparisons are provided in Table S2. n indicates sample sizes for Field, Lab and Urban colony-collection environments, respectively. Bar plots represent means ± standard deviation.

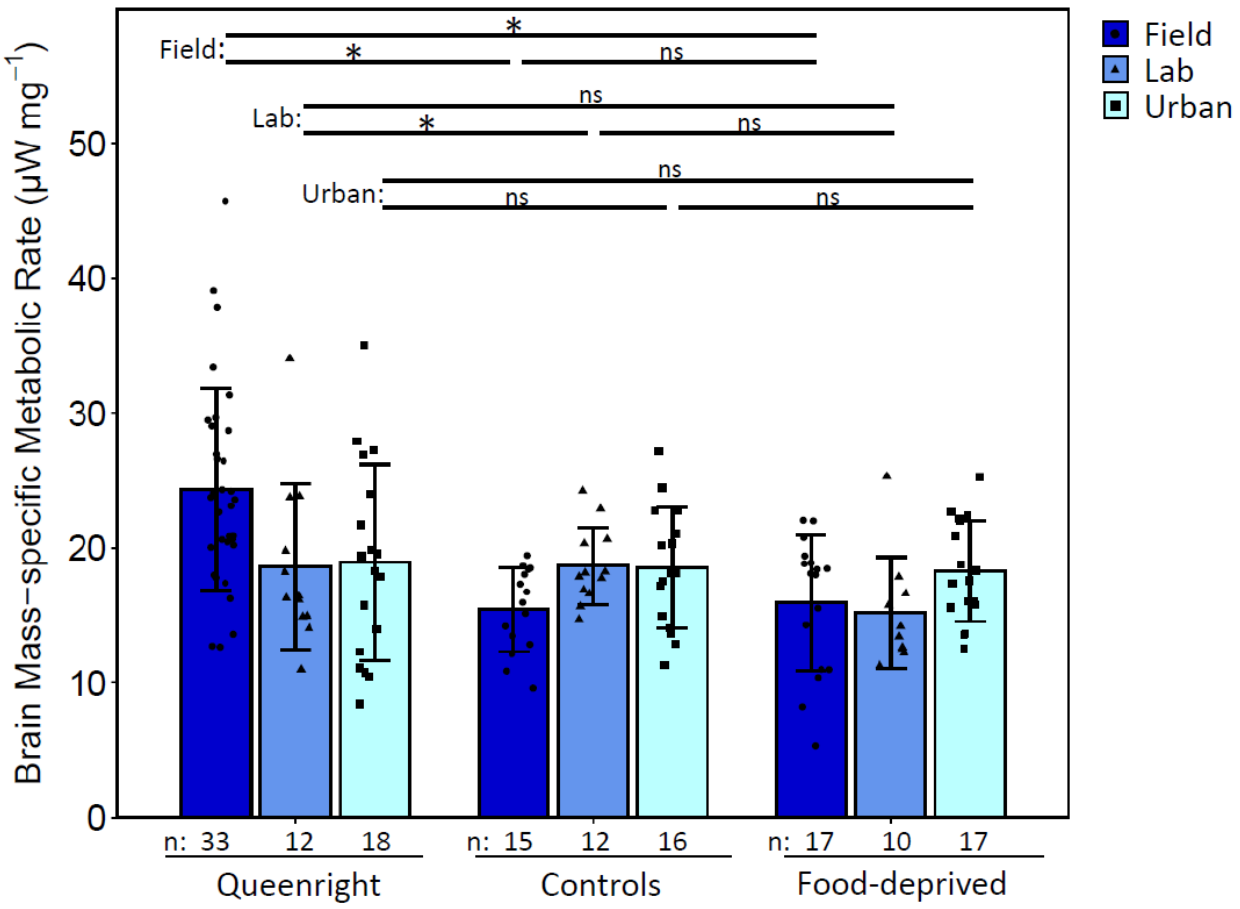


Fig S6. Colony-collection environment did not have a significant effect on brain mass-specific metabolic rate (Table S10, $F = 0.81$, $p = 0.51$), but there was a significant treatment-environment interaction effect (Table S10, $F = 6.31$, $p < 0.01$). For workers collected from field colonies, queenright workers exhibited higher brain mass-specific metabolic rates than controls ($T = 3.84$, $p < 0.01$) and food-deprived workers ($T = 4.57$, $p < 0.01$). For workers collected from lab colonies, controls had higher brain mass-specific metabolic rate than queenright workers ($T = 2.51$, $p = 0.04$). Statistics of all pairwise comparisons are provided in Table S3. n indicates sample sizes for Field, Lab and Urban colony-collection environments, respectively. Bar plots represent means ± standard deviation.

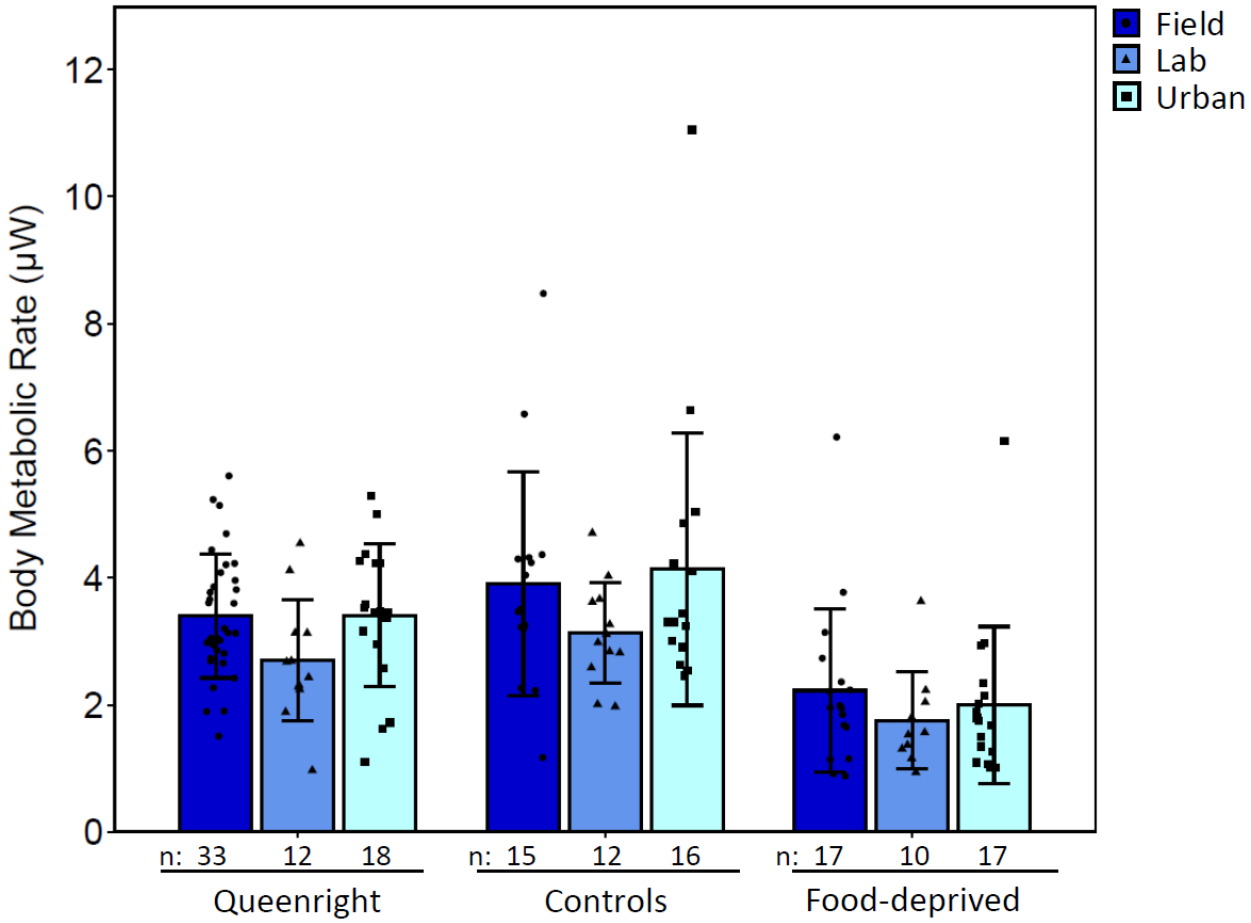


Fig S7. Colony-collection environment did not have a significant effect on body metabolic rate (Table S11, F = 0.03, p = 0.97) and there was no significant treatment-environment interaction effect (Table S11, F = 0.93, p = 0.45). n indicates sample sizes for Field, Lab and Urban colony-collection environments, respectively. Bar plots represent means ± standard deviation.

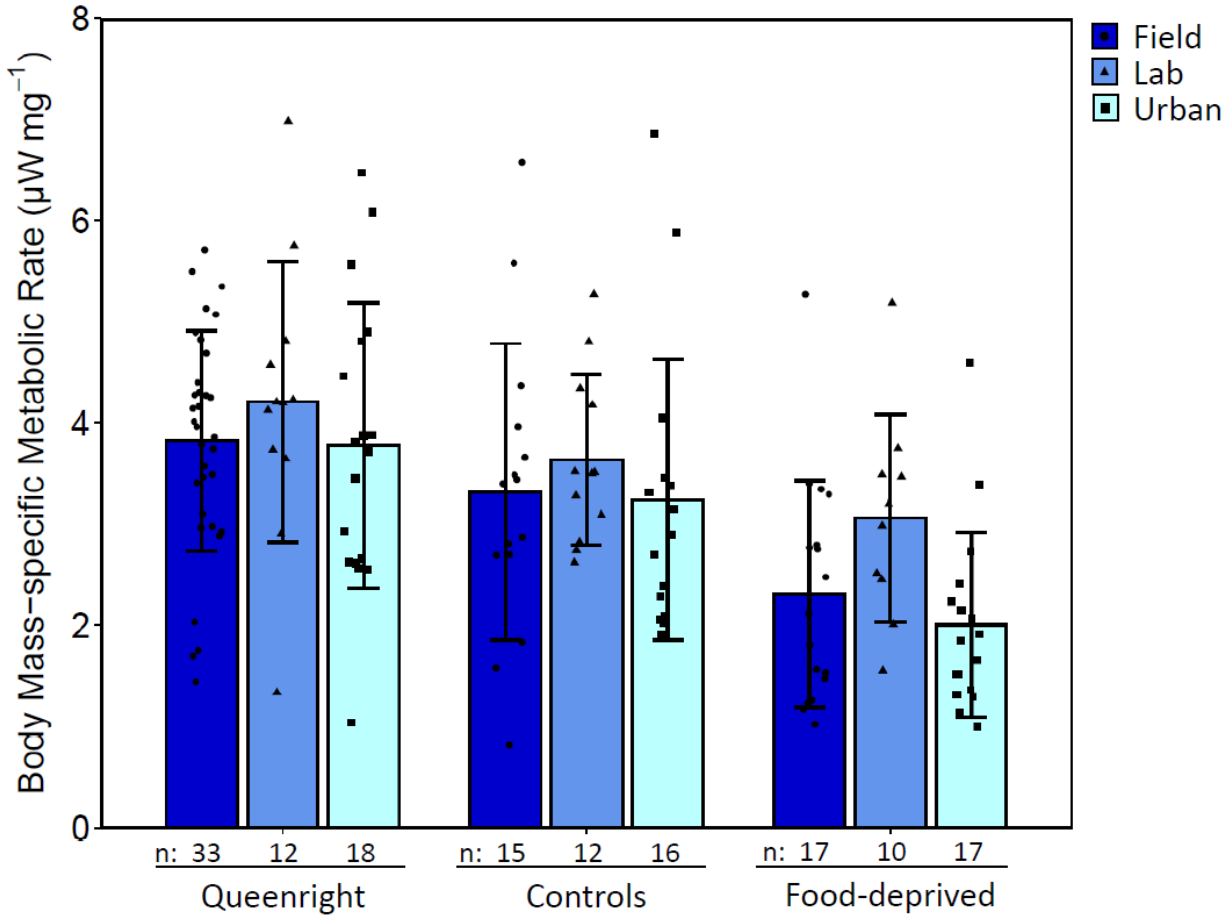


Fig S8. Colony-collection environment did not have a significant effect on body mass-specific metabolic rate (Table S12, F = 0.96, p = 0.44) and there was no significant treatment-environment interaction effect (Table S12, F = 0.49, p = 0.74). n indicates sample sizes for Field, Lab, and Urban environments, respectively. Bar plots represent means ± standard deviation.

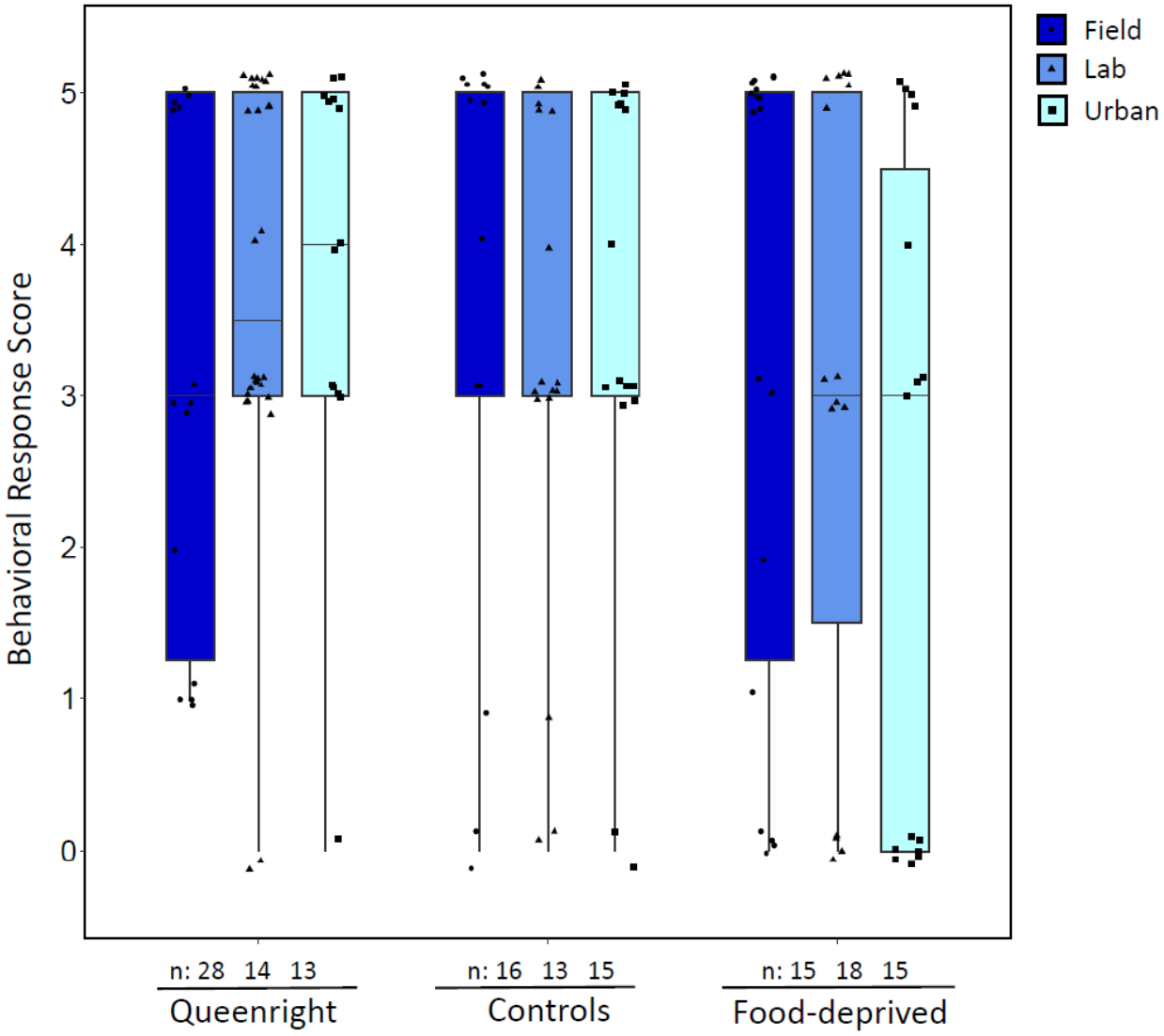


Fig S9. Colony-collection environment did not significantly impact worker behavioral responses (Kruskal-Wallis: $\chi^2 = 0.345$, $p = 0.84$). Statistics of pairwise comparisons are provided in Table S3. Box plots represent median ± 150% of the interquartile range.

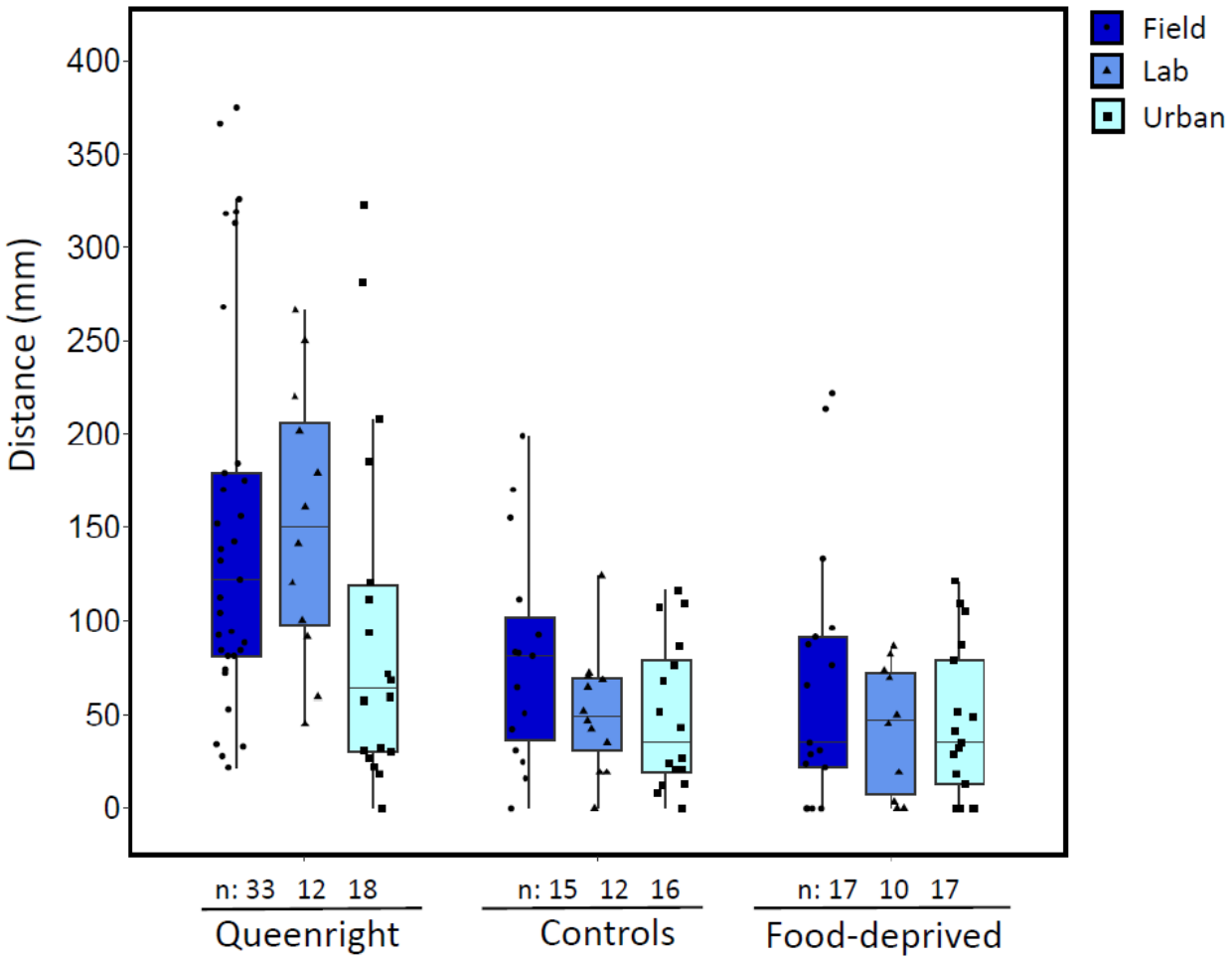


Fig S10. Colony-collection environment had a significant effect on worker movement during respirometry recordings (Table S13, $\chi^2 = 8.76$, $p = 0.01$). Workers from field colonies exhibited significantly greater movement distances than workers from urban colonies ($Z = 2.92$, $p < 0.01$). There were no significant Treatment-Environment interaction effects (Table S13, $\chi^2 = 2.63$, $p = 0.62$). n indicates sample sizes for Field, Lab and Urban colony-collection environments, respectively. Box plots represent medians ± 150% of the interquartile range.

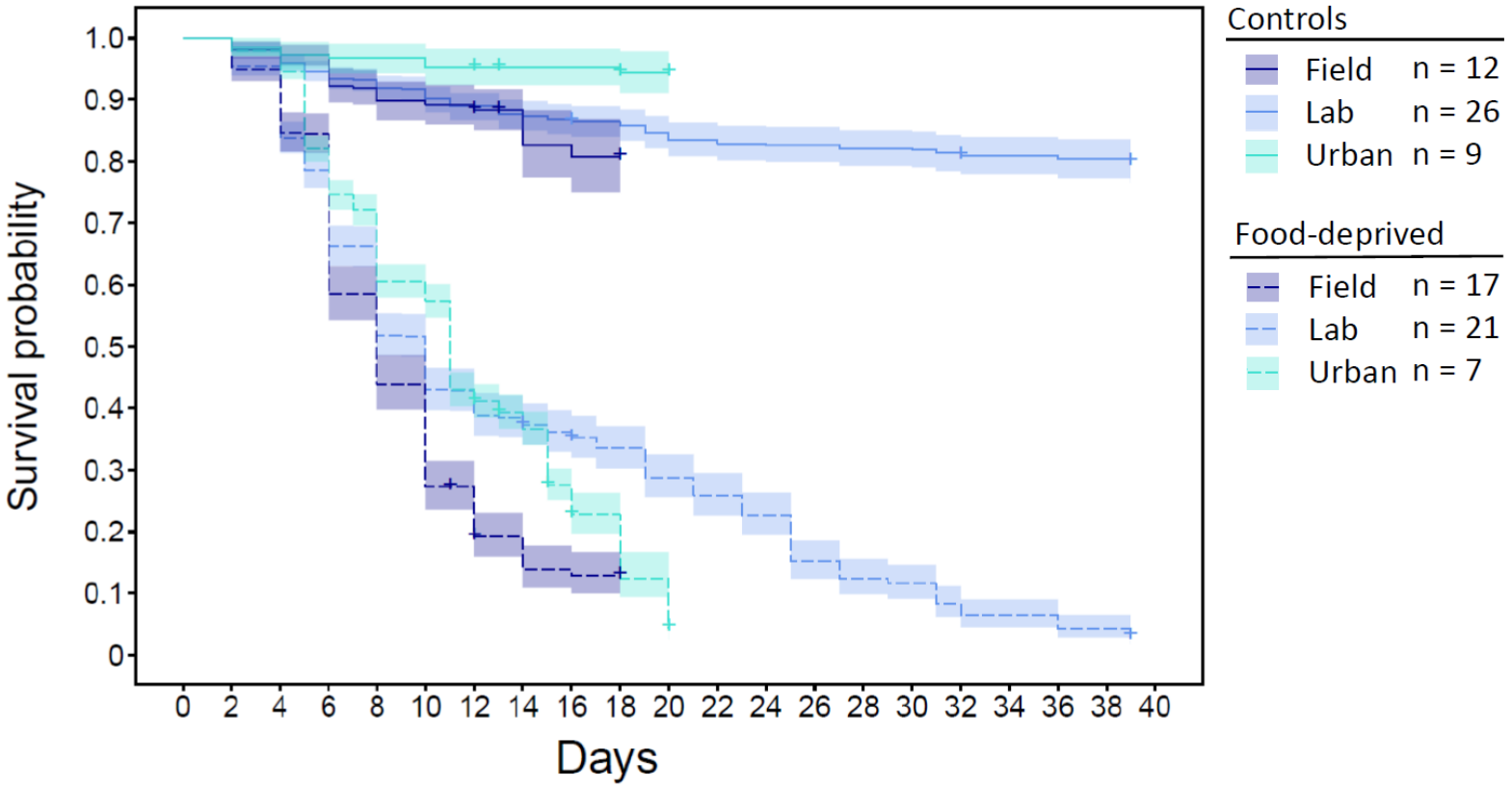


Fig S11. Survival probability curves of workers in control or food-deprived subcolonies from colony-collection environments (Field, Laboratory, Urban). Opaque bands are 95% confidence intervals.

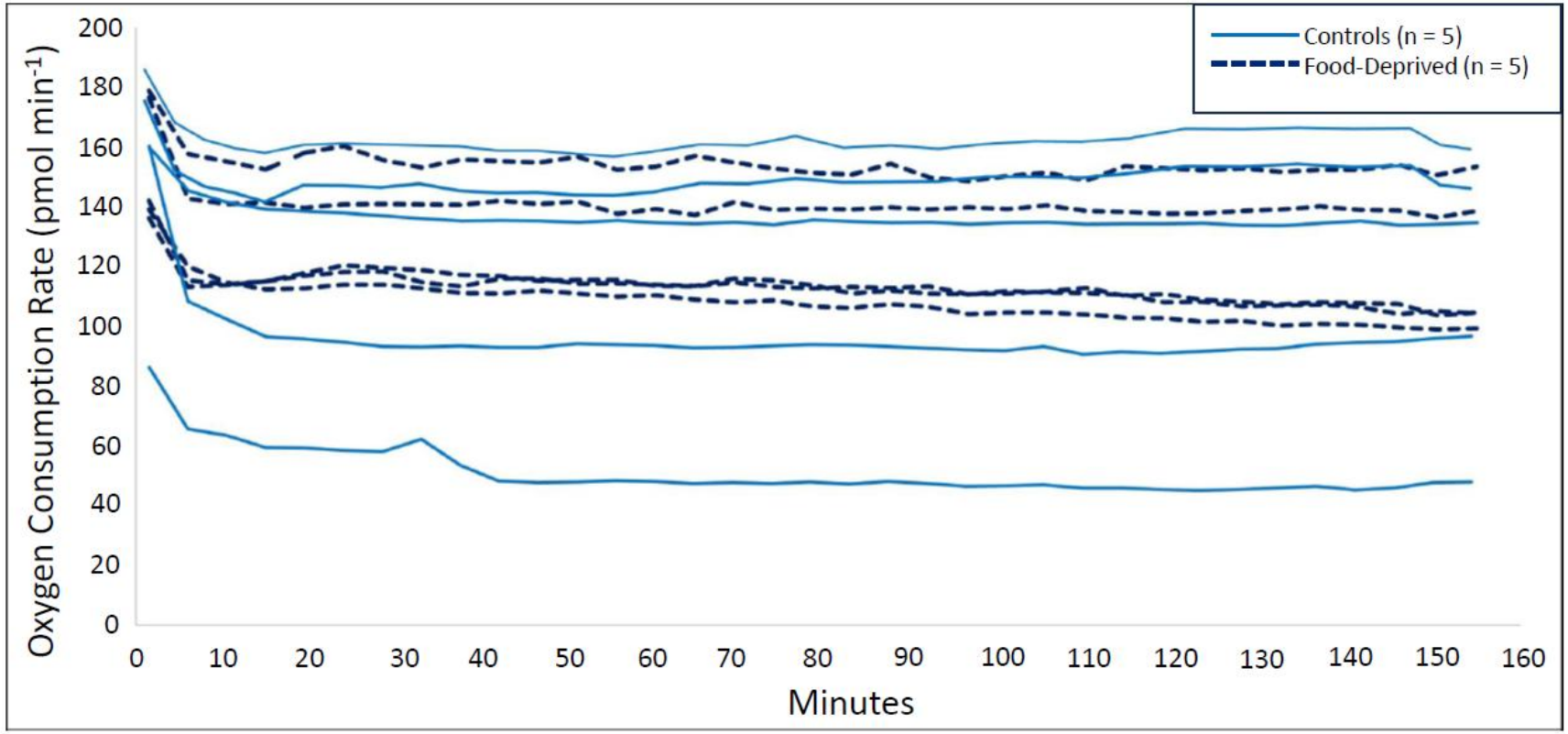
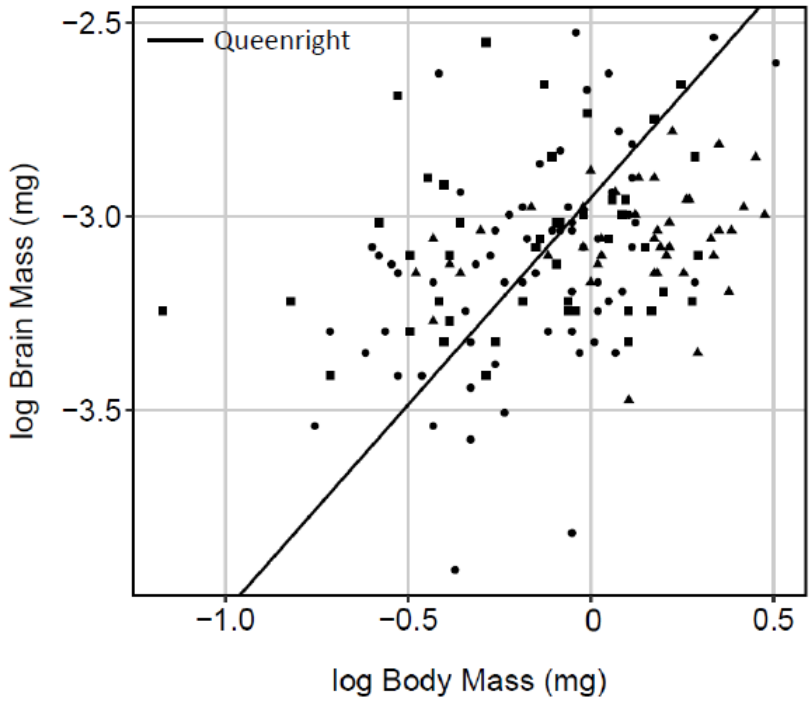
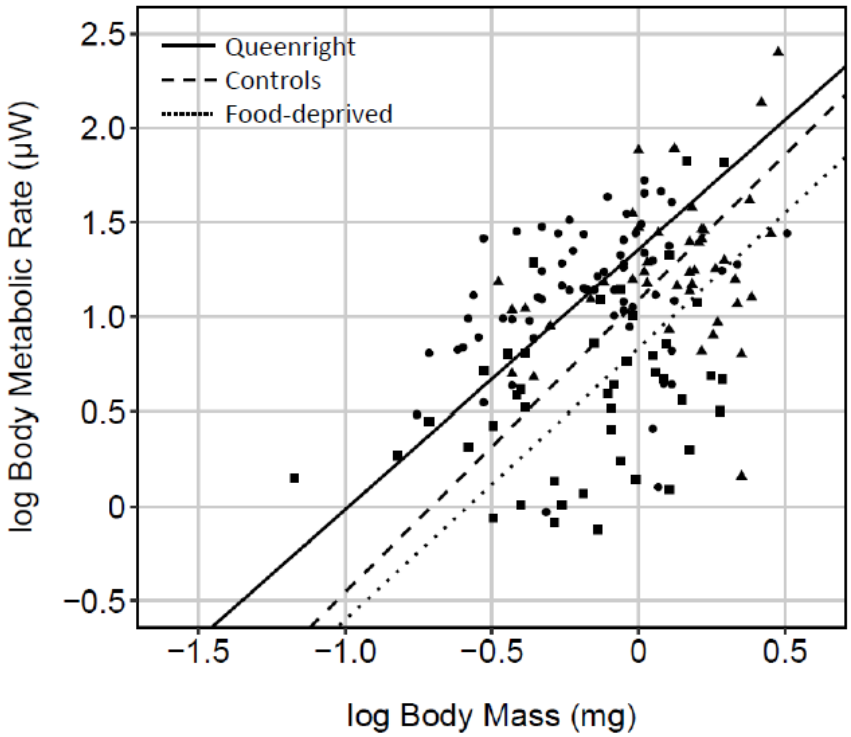


Fig S12. Representative traces of brain oxygen consumption rates (OCR) of controls and food-deprived workers illustrating the stability of recordings. Each trace represents the recording of a single brain over 150 minutes. n indicates the number of traces displayed in the figure from each treatment. Recorded OCR was typically elevated during the first several measurements prior to stabilizing. Therefore, OCR values were sampled between 20-50 minutes from the start of the assay.



	Queenright (n=63) ●	Controls (n=43) ▲	Food-deprived (n=44) ■
Slope ± 95% CI	1.07 ± 0.26	0.52 ±0.18	0.67 ± 0.23
Y-intercept ± 95% CI	-2.95 ± 0.08	-3.11 ± 0.05	-2.96 ± 0.09
Slope p value	<0.01	0.07	0.16
r^2	0.203	0.097	0.047

Fig S13. Standard major axis regression of log brain mass scaled to log body mass. A p value < 0.05 indicates that the scaling slope was only significant for workers sampled directly from queenright colonies. The regression line represents the slope only for queenright workers.

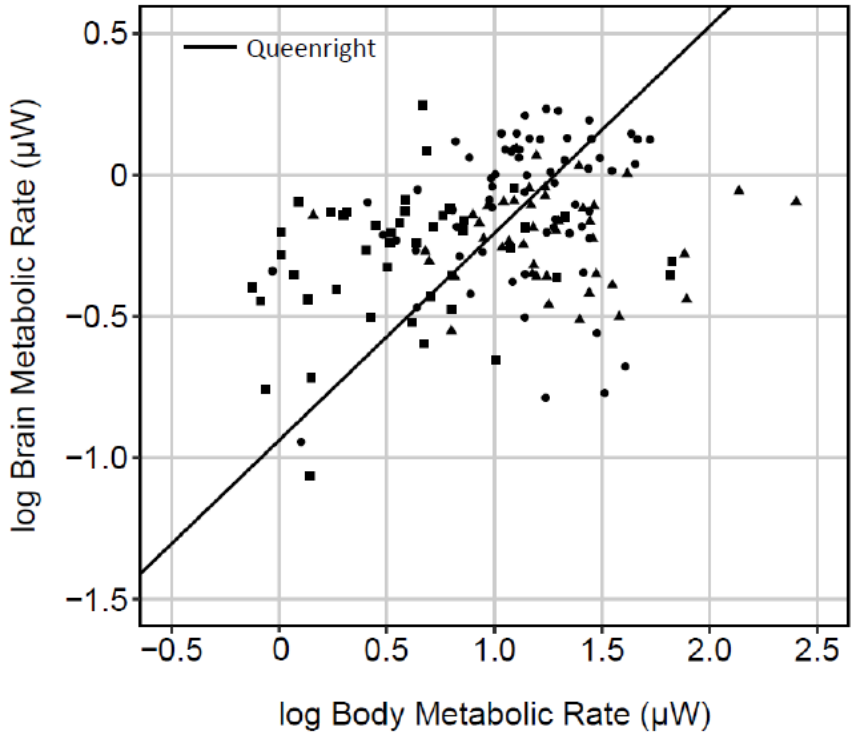


	Queenright (n=63) ●	Controls (n=43) ▲	Food-deprived (n=44) ■
Slope ± 95% CI	1.37 ± 0.30	1.44 ± 0.40	1.44 ± 0.35
Y-intercept ± 95% CI	1.35 ± 0.12	1.09 ± 0.15	0.83 ± 0.17
slope p value	0.02	0.05	< 0.01
r^2	0.087	0.091	0.157

Slope Comparisons	r statistic	p value	Slope difference?
Queenright - Food-deprived	0.054	0.99	No
Queenright - Controls	0.363	0.91	No
Controls - Food-deprived	0.125	0.98	No

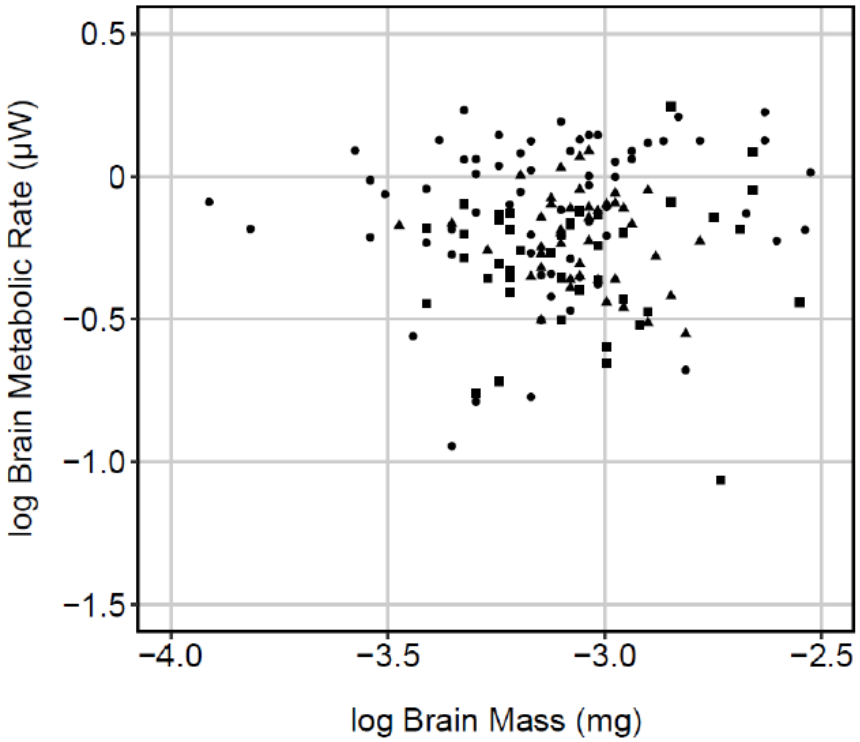
Elevation Comparisons	W statistic	p value	Grade shift?
Queenright - Food-deprived	32.42	< 0.01	Yes
Queenright - Controls	7.90	0.01	Yes
Controls - Food-deprived	5.00	0.07	No

Fig S14. Standard major axis regression of log body metabolic rate scaled to log body mass. p values < 0.05 indicates that queenright workers and food-deprived workers exhibit significant scaling slopes. Pairwise comparisons indicate that queenright workers differed from food-deprived workers ($W = 32.42$, $p < 0.01$) and controls ($W = 7.90$, $p = 0.01$) in elevation.



	Queenright (n=63) ●	Controls (n=43) ▲	Food-deprived (n=44) ■
Slope ± 95% CI	0.73 ± 0.20	0.41 ± 0.11	0.50 ± 0.37
Y-intercept ± 95% CI	-0.94 ± 0.22	-0.73 ± 0.18	-0.60 ± 0.12
slope p value	0.04	0.89	0.16
r^2	0.069	< 0.001	0.046

Fig S15. Standard major axis regression of log brain metabolic rate scaled to log body metabolic rate . A p value < 0.05 indicates that queenright workers exhibit a significant scaling slope. The regression line represents the slope only for queenright workers.



	Queenright (n=63) ●	Controls (n=43) ▲	Food-deprived (n=44) ■
Slope ± 95% CI	0.94 ± 0.21	-1.23 ± 0.44	1.08 ± 0.39
Y-intercept ± 95% CI	2.83 ± 0.75	-3.96 ± 1.16	3.01 ± 1.03
slope p value	0.24	0.18	0.74
r^2	0.022	0.043	0.003

Fig S16. Standard major axis regression of log brain metabolic rate scaled to log brain mass. p-values > 0.05 indicate that scaling was not significant in any treatment.

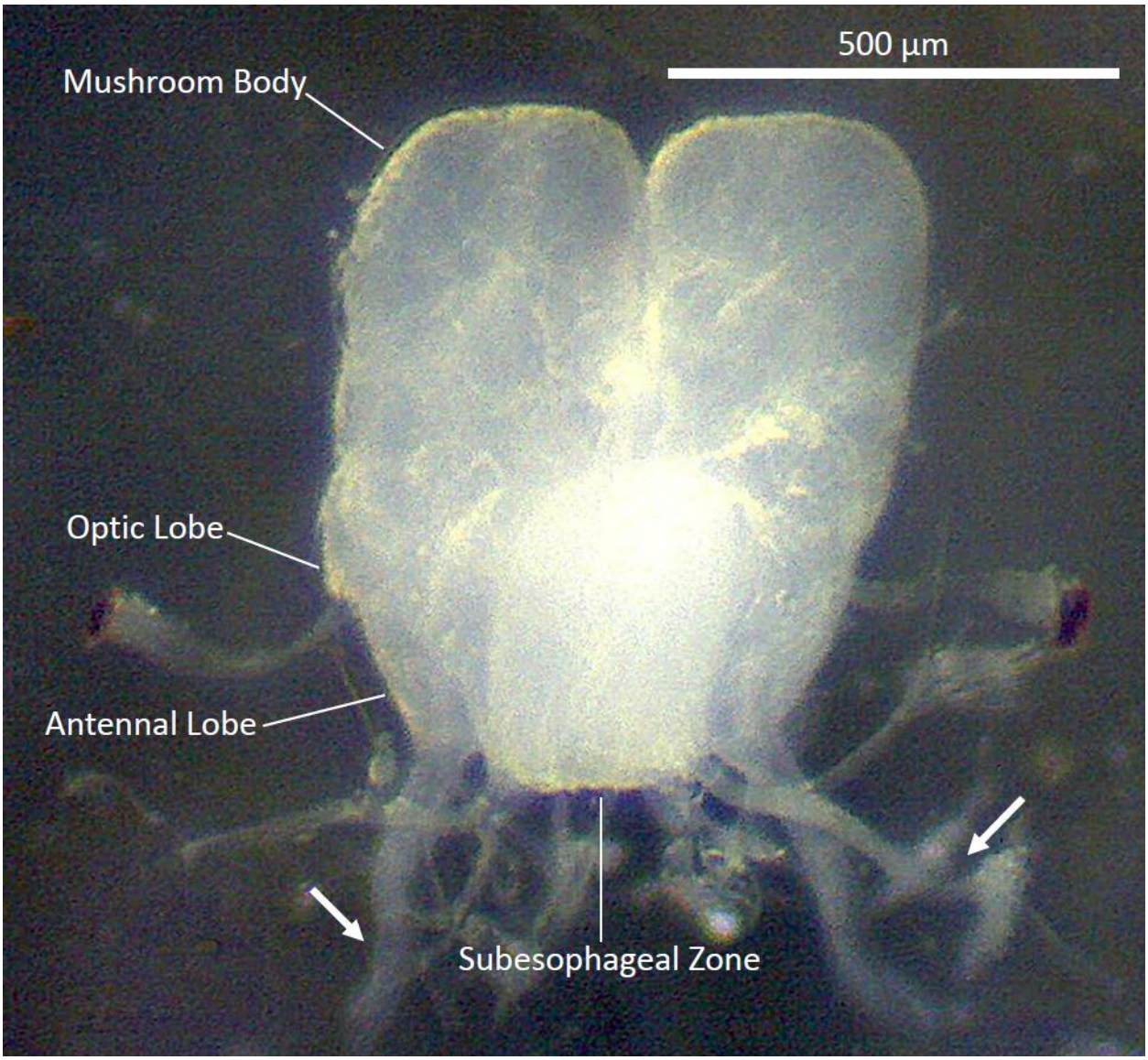


Fig S17. Representative image of a whole, undamaged *T. immigrans* brain, including intact optic nerves terminating in the retina (reddish-brown, left and right). Note the antennal and subesophageal nerves cut on axons (white arrows) distally from the brain to avoid tissue injury. Brain visualization from a Celestron Digital Microscope Imager Model 44421.

Supplementary Tables

Table S1. Summary of worker brain mass, body mass, and metabolic rates across treatments. The first row of each response variable provides the mean ± standard deviation. The second row provides the median (first and third quartiles bracketed).

Response Variable	**Queenright**	**Controls**	**Food-deprived**
Brain Mass (mg)	0.045 ± 0.013 0.044 [0.037, 0.050]	0.048 ± 0.006 0.047 [0.044, 0.051]	0.048 ± 0.01 0.046 [0.039, 0.052]
Brain Metabolic Rate (μW)	0.92 ± 0.21 0.95 [0.80, 1.09]	0.81 ± 0.13 0.82 [0.70, 0.90]	0.76 ± 0.17 0.78 [0.66, 0.86]
Brain Mass-specific Metabolic Rate ($\mu W\ mg^{-1}$)	21.0 ±7.60 20.61 [16.27, 26.53]	17.5 ± 3.85 17.74 [14.84, 19.80]	16.7 ± 4.46 17.02 [13.52, 19.02]
Body Mass (mg)	0.88 ± 0.23 0.89 [0.72, 1.02]	1.14 ± 0.26 1.19 [1.00, 1.31]	0.89 ± 0.26 0.91 [0.68, 1.09]
Body Metabolic Rate (μW)	3.27 ±1.03 3.13 [2.67, 4.02]	3.78 ± 1.74 3.31 [2.87, 4.23]	2.03 ± 1.15 1.80 [1.30, 2.23]
Body Mass-specific Metabolic Rate ($\mu W\ mg^{-1}$)	3.88 ± 1.24 3.88 [2.96, 4.63]	3.37 ± 1.27 3.31 [2.69, 3.81]	2.36 ± 1.08 2.12 [1.51, 3.03]
Relative Brain Mass (%)	4.54 ± 1.28 5.22 [4.33, 6.24]	4.75 ± 0.61 4.00 [3.60, 4.96]	4.75 ± 1.08 5.36 [4.39, 6.65]
Relative Brain Metabolic Rate (%)	29.9 ± 11.9 30.02 [22.59, 36.67]	24.4 ± 0.1 25.08 [19.22, 30.63]	45.0 ± 18.3 42.80 [31.80, 55.05]

Table S2. Summary statistics of pairwise comparisons of estimated marginal means of brain metabolic rate between treatments within each colony-collection environment. Comparisons are Q - C (Queenright - Controls), Q - FD (Queenright - Food-deprived), and C - FD (Controls - Food-deprived).

Environment	Contrast	Estimate	Std. Error	T statistic	p value
Field Colonies	Q - C	0.195	0.085	T = 2.30	0.06
	Q - FD	0.223	0.067	T = 3.32	< 0.01
	C - FD	0.028	0.061	T = 0.46	0.89
Lab Colonies	Q - C	0.147	0.068	T = 2.15	0.08
	Q - FD	0.104	0.083	T = 1.26	0.42
	C - FD	0.042	0.077	T = 0.55	0.85
Urban Colonies	Q - C	0.079	0.072	T = 1.11	0.51
	Q - FD	0.074	0.061	T = 1.20	0.46
	C - FD	0.006	0.062	T = 0.10	0.99

Table S3. Summary statistics of pairwise comparisons of estimated marginal means of brain mass-specific metabolic rate between treatments within each colony-collection environment. Comparisons are Q - C (Queenright - Controls), Q - FD (Queenright - Food-deprived), and C - FD (Controls - Food-deprived).

Environment	Contrast	Estimate	Std. Error	T statistic	p value
Field Colonies	Q - C	0.420	0.109	T = 3.84	< 0.01
	Q - FD	0.405	0.089	T = 4.57	< 0.01
	C - FD	0.015	0.110	T = 0.14	0.99
Lab Colonies	Q - C	0.311	0.124	T = 2.51	0.04
	Q - FD	0.084	0.150	T = 0.56	0.84
	C - FD	0.227	0.141	T = 1.61	0.25
Urban Colonies	Q - C	0.020	0.120	T = 0.17	0.99
	Q - FD	0.011	0.097	T = 0.11	0.99
	C - FD	0.031	0.116	T = 0.27	0.96

Table S4. Summary ofDunn test results for pairwise comparisons of behavioral response scores within each environment. The Bonferroni method was used to correct for multiple comparisons.

Environment	Contrast	Z statistic	p value
Field	Q - C	0.973	0.99
	Q - FD	0.983	0.98
	C - FD	0.026	1.00
Lab	Q - C	0.672	1.00
	Q - FD	0.456	1.00
	C - FD	0.265	1.00
Urban	Q - C	0.558	1.00
	Q - FD	1.947	0.15
	C - FD	1.442	0.45

Table S5. Summary of general linear mixed model diagnostics, fixed effects and interactions using an inverse-Gaussian distribution of brain mass as a dependent variable. AIC, KS Test and model dispersion were used to evaluate model fit. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, or Environment), Body Mass (BoMa). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. The model was simplified by sequentially dropping non-significant terms; each subsequent model is described in a separate row of the table. The simplest model for which assumptions were met is indicated with an asterisk (*) and used to interpret results (Tables 2-3 in the main manuscript).

Model	**AIC**	**KS Test**	**Dispersion**	**Fixed Effects**
1	-948	$D = 0.08$ $p = 0.26$	$D = 0.89$ $p = 0.42$	Treatment: $\eta^2 = 0.03$, $\chi^2 = 0.63$, $p = 0.73$ Environment: $\eta^2 = 0.03$, $\chi^2 = 1.84$, $p = 0.40$ Body Mass: $\eta^2 = 0.10$, $\chi^2 = 13.02$, $p < 0.01$ T-E: $\eta^2 = 0.03$, $\chi^2 = 4.72$, $p = 0.32$ T-BoMa: $\eta^2 = 0.05$, $\chi^2 = 7.74$, $p = 0.02$
2*	-952	$D = 0.07$ $p = 0.47$	$D = 0.91$ $p = 0.64$	Treatment: $\eta^2 = 0.03$, $\chi^2 = 0.87$, $p = 0.65$ Environment: $\eta^2 = 0.02$, $\chi^2 = 1.50$, $p = 0.47$ Body Mass: $\eta^2 = 0.10$, $\chi^2 = 13.62$, $p < 0.01$ T-BoMa: $\eta^2 = 0.05$, $\chi^2 = 7.09$, $p = 0.03$

Table S6. Summary of general linear mixed model diagnostics, fixed effects and interactions using a Gaussian distribution of body mass as the dependent variable. AIC, a KS Test, model dispersion, a Levene's Test and a Shapiro-Wilks Test were used to evaluate model fit. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, or Environment). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. The model was simplified by sequentially dropping non-significant terms; each subsequent model is described in a separate row of the table. The simplest model for which assumptions were met is indicated with an asterisk (*) and used to interpret results (Tables 2-3 in the main manuscript).

Model	**AIC**	**KS Test**	**Dispersion**	**Levene's Test**	**Shapiro-Wilk Test**	**Fixed Effects**
1	-13	D = 0.08 p = 0.27	D = 0.75 p = 0.24	Environment: F = 1.65, p = 0.20 Treatment: F = 0.67. p = 0.52	W = 0.99 p = 0.35	Treatment: $\eta^2 = 0.31$, F = 28.94, p < 0.01 Environment: $\eta^2 = 0.28$, F = 2.12, p = 0.17 T-E: $\eta^2 = 0.01$, F = 0.46, p = 0.77
2*	-40	D = 0.07 p = 0.49	D = 0.78 p = 0.34	Environment: F = 1.86, p = 0.16 Treatment: F = 0.64, p = 0.53	W = 0.99 p = 0.22	Treatment: $\eta^2 = 0.30$, F = 30.48, p < 0.01 Environment: $\eta^2 = 0.27$, F = 2.14, p = 0.16 No interaction

Table S7. Summary of general linear mixed model diagnostics of fixed effects and interactions using an inverse Gaussian distribution of relative brain mass as the dependent variable. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, or Environment). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. The model was simplified by sequentially dropping non-significant terms; each subsequent model is described in a separate row of the table. The simplest model for which assumptions were met is indicated with an asterisk (*) and used to interpret results (Tables 2-3 in the main manuscript).

Model	**AIC**	**KS Test**	**Dispersion**	**Fixed Effects**
1	-871	$D = 0.04$ $p = 0.97$	$D = 1.03$ $p = 0.73$	Treatment: $\eta^2 = 0.17$, $\chi^2 = 30.82$, $p < 0.01$ Environment: $\eta^2 = 0.17$, $\chi^2 = 30.62$, $p < 0.01$ T-E: $\eta^2 = 0.05$, $\chi^2 = 7.33$, $p = 0.12$
2*	-870	$D = 0.05$ $p = 0.79$	$D = 1.15$ $p = 0.34$	Treatment: $\eta^2 = 0.18$, $\chi^2 = 33.25$, $p < 0.01$ Environment: $\eta^2 = 0.17$, $\chi^2 = 30.63$, $p < 0.01$ No Interaction included

Table S8. Summary of general linear mixed model diagnostics of fixed effects and interactions using a Gaussian distribution of relative brain metabolic rate as the dependent variable. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, or Environment), Body Mass (BoMa), Brain Mass (BrMa). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. No data transformation met distribution assumptions, including after models were simplified by dropping non-significant terms (models 4-5). Therefore, the model with the lowest AIC value (model 1) was used to interpret results, indicated with an asterisk (*).

Model	AIC	KS Test	Dispersion	Levene's Test	Shapiro-Wilk Test	Fixed Effects
1 (square root-transformed)*						
	-150	D = 0.09 p = 0.16	D = 0.83 p = 0.26	Environment: F = 0.92, p = 0.40 Treatment: F = 6.56, p < 0.01	W = 0.99 p = 0.67	Treatment: η^2 = 0.16, F = 10.94, p < 0.01 Environment: η^2 = 0.04, F = 0.20, p = 0.82 BoMa: η^2 = 0.04, F = 5.79, p = 0.02 BrMa: η^2 < 0.01, F = 0.62, p = 0.43 T-E: η^2 = 0.06, F = 1.70, p = 0.15 T-BoMa: η^2 = 0.02, F = 1.54, p = 0.22
2 (data not transformed)						
	-103	D = 0.10 p = 0.08	D = 0.82 p = 0.26	Environment: F = 0.70, p = 0.50 Treatment: F = 12.49, p < 0.01	W = 0.97 p < 0.01	Treatment: η^2 = 0.18, F = 12.22, p < 0.01 Environment: η^2 = 0.04, F = 0.19, p = 0.83 BoMa: η^2 = 0.03, F = 3.92, p = 0.050 BrMa: η^2 < 0.01, F = 1.08, p = 0.30 T-E: η^2 = 0.08, F = 2.14, p = 0.08 T-BoMa: η^2 = 0.03, F = 1.87, p = 0.16
3 (logit-transformed)						
	330	D = 0.10 p = 0.12	D = 0.83 p = 0.26	Environment: F = 0.82, p = 0.44 Treatment: F = 3.98, p = 0.02	W = 0.99 p = 0.13	Treatment: η^2 = 0.15, F = 10.48, p < 0.01 Environment: η^2 = 0.05, F = 0.21, p = 0.82 BoMa: η^2 = 0.04, F = 5.06, p = 0.03 BrMa: η^2 < 0.01, F = 0.85, p = 0.36 T-E: η^2 = 0.07, F = 1.95, p = 0.11 T-BoMa: η^2 = 0.02, F = 1.56, p = 0.21

Table S8 (continued)

Model	AIC	KS Test	Dispersion	Levene's Test	Shapiro-Wilk Test	Fixed Effects
4 (log-transformed)						
	206	D = 0.09 p = 0.20	D = 0.84 p = 0.27	Environment: F = 1.04, p = 0.36 Treatment: F = 2.24, p = 0.11	W = 0.99 p = 0.12	Treatment : $\eta^2 = 0.13$, F = 8.77, p < 0.01 Environment: $\eta^2 = 0.05$, F = 0.22, p = 0.81 BoMa: $\eta^2 = 0.05$, F = 7.39, p < 0.01 BrMa : $\eta^2 < 0.01$, F = 0.27, p = 0.60 T-E : $\eta^2 = 0.05$, F = 1.37, p = 0.25 T-BoMa: $\eta^2 = 0.02$, F = 1.11, p = 0.33
5 (log-transformed)						
	206	D = 0.09 p = 0.26	D = 0.86 p = 0.33	Environment: F = 1.59, p = 0.21 Treatment: F = 2.20, p = 0.11	W = 0.98 p = 0.03	Treatment : $\eta^2 = 0.17$ F = 11.80, p < 0.01 Environment: $\eta^2 = 0.04$, F = 0.18, p = 0.83 BoMa: $\eta^2 = 0.05$, F = 6.73, p = 0.01 BrMa : $\eta^2 < 0.01$, F = 0.42, p = 0.52 T-E : $\eta^2 = 0.03$, F = 0.91, p = 0.46

Table S9. Summary of general linear mixed model diagnostics of fixed effects and interactions using a Gaussian distribution of brain metabolic rate as the dependent variable. AIC, a KS Test, model dispersion, a Levene's Test and a Shapiro-Wilks Test were used to evaluate model fit. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, Environment), Body Metabolic Rate (BoMr), Body Mass (BoMa), Brain Mass (BrMa). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. Models were simplified by sequentially dropping non-significant terms; each subsequent model is described in a separate row of the table.

Model	AIC	KS Test	Dispersion	Levene's Test	Shapiro-Wilk Test	Fixed Effects
1	-35	D = 0.09 p = 0.16	D = 0.73 p = 0.10	Environment: F = 1.15, p = 0.32 Treatment: F = 0.92. p = 0.40	W = 0.99 p = 0.42	Treatment: $\eta^2 < 0.01$, F = 0.50, p = 0.61 Environment: $\eta^2 = 0.04$, F = 0.23, p = 0.80 BoMr: $\eta^2 < 0.01$, F = 1.29, p = 0.26 BoMa: $\eta^2 < 0.01$, F < 0.01, p = 0.95 BrMa: $\eta^2 = 0.01$, F = 1.87, p = 0.17 T-E: $\eta^2 = 0.11$, F = 3.58, p < 0.01 T-BoMr: $\eta^2 = 0.02$, F = 1.31, p = 0.27 T-BoMa: $\eta^2 = 0.07$, F = 5.13, p < 0.01
2	-50	D = 0.09 p = 0.20	D = 0.74 p = 0.13	Environment: F = 2.00, p = 0.14 Treatment: F = 1.23, p = 0.29	W = 0.99 p = 0.46	Treatment: $\eta^2 = 0.02$, F = 1.03, p = 0.36 Environment: $\eta^2 = 0.05$, F = 0.26, p = 0.77 BoMr: $\eta^2 < 0.01$, F = 0.65, p = 0.42 BoMa: $\eta^2 < 0.01$, F < 0.01, p = 0.95 BrMa: $\eta^2 = 0.02$, F = 2.52, p = 0.11 T-E: $\eta^2 = 0.11$, F = 3.73, p = 0.01 T-BoMa : $\eta^2 = 0.06$, F = 4.57, p = 0.01
3	-58	D = 0.09 p = 0.15	D = 0.75 p = 0.14	Environment: F = 2.20, p = 0.11 Treatment: F = 1.20, p = 0.30	W = 1.20 p = 0.30	Treatment: $\eta^2 = 0.02$, F = 1.38, p = 0.25 Environment: $\eta^2 = 0.05$, F = 0.25, p = 0.78 BoMa: $\eta^2 < 0.01$, F = 0.08, p = 0.77 BrMa: $\eta^2 = 0.02$, F = 2.73, p = 0.11 T-E: $\eta^2 = 0.11$, F = 3.88, p = 0.01 T-BoMa: $\eta^2 = 0.07$, F = 4.96, p = 0.01

Table S9 (continued)

Model	AIC	KS Test	Dispersion	Levene's Test	Shapiro-Wilk Test	Fixed Effects
4*	-55	D = 0.09 p = 0.18	D = 0.73 p = 0.12	Environment: F = 1.99, p = 0.14 Treatment: F = 1.59, p = 0.21	W = 0.99 p = 0.81	Treatment: η^2 = 0.02, F = 1.17, p = 0.31 Environment: η^2 = 0.05, F = 0.26, p = 0.77 BoMa: η^2 < 0.01, F = 0.52, p = 0.47 T-E: η^2 = 0.10, F = 3.42, p = 0.01 T-BoMa: η^2 = 0.06, F = 4.37, p = 0.01

Table S10. Summary of general linear mixed model diagnostics of fixed effects and interactions using a Gaussian distribution with log-transformed brain mass-specific metabolic rate as the dependent variable. AIC, a KS Test, model dispersion, a Levene's Test and a Shapiro-Wilks Test were used to evaluate model fit. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, Environment), Body Mass (BoMa), Body Metabolic Rate (BoMr). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. The model was simplified by sequentially dropping non-significant terms; each subsequent model is described in a separate row of the table. The simplest model for which assumptions were met is indicated with an asterisk (*) and used to interpret results.

Model	AIC	KS Test	Dispersion	Levene's Test	Shapiro-Wilk Test	Fixed Effects
1	123	D = 0.08 p = 0.37	D = 0.89 p = 0.30	Environment: F = 2.35, p = 0.10 Treatment: F = 0.97, p = 0.38	W = 0.99 p = 0.18	Treatment: η^2 = 0.01, F = 0.61, p = 0.55 Environment: η^2 = 0.27, F = 0.79, p = 0.51 BoMa: η^2 = 0.04, F = 4.45, p = 0.04 BoMr: η^2 < 0.01, F = 0.01, p = 0.92 T-E: η^2 = 0.23, F = 6.02, p < 0.01 T-BoMa: η^2 = 0.10, F = 6.82, p < 0.01 T-BoMr: η^2 < 0.01, F = 0.53, p = 0.59
2	109	D = 0.07 p = 0.49	D = 0.91 p = 0.42	Environment: F = 2.62, p = 0.08 Treatment: F = 1.03, p = 0.36	W = 0.98 p = 0.06	Treatment: η^2 = 0.02, F = 1.19, p = 0.31 Environment: η^2 = 0.31, F = 0.82, p = 0.51 BoMa: η^2 = 0.04, F = 3.88, p = 0.05 BoMr: η^2 < 0.01, F = 0.05, p = 0.83

						T-E: $\eta^2 = 0.24$, $F = 6.32$, $p < 0.01$ T-BoMa: $\eta^2 = 0.11$, $F = 8.01$, $p < 0.01$
3*	101	$D = 0.07$ $p = 0.49$	$D = 0.91$ $p = 0.47$	Environment: $F = 2.53$, $p = 0.08$ Treatment: $F = 1.12$, $p = 0.33$	$W = 0.98$ $p = 0.06$	Treatment: $\eta^2 = 0.02$, $F = 1.20$, $p = 0.31$ Environment: $\eta^2 = 0.30$, $F = 0.81$, $p = 0.51$ BoMa: $\eta^2 = 0.04$, $F = 4.56$, $p = 0.04$ T-E: $\eta^2 = 0.24$, $F = 6.31$, $p < 0.01$ T-BoMa: $\eta^2 = 0.11$, $F = 8.04$, $p < 0.01$

Table S11. Summary of general linear mixed model diagnostics of fixed effects and interactions using a Gaussian distribution of body metabolic rate as the dependent variable. AIC, a KS Test, model dispersion, a Levene's Test and a Shapiro-Wilks Test were used to evaluate model fit. No data transformations were sufficient to fit a gaussian distribution, but a square-root transformation resulted in a model with the lowest AIC score. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, separated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, Environment), Body Mass (BoMa). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. No data transformations met distribution assumptions, including after simplifying models by sequentially dropping non-significant terms (models 3-4). Therefore, the model with the lowest AIC value (model 1) was used to interpret results, indicated with an asterisk (Tables 2-3 in the main manuscript).

Model	**AIC**	**KS Test**	**Dispersion**	**Levene's Test**	**Shapiro-Wilk Test**	**Fixed Effects**
1 (square root-transformed)*						
	152	$D = 0.08$ $p = 0.29$	$D = 0.88$ $p = 0.35$	Environment: $F = 2.53$, $p = 0.08$ Treatment: $F = 0.38$. $p = 0.68$	$W = 0.97$ $p < 0.01$	Treatment: $\eta^2 = 0.16$, $F = 10.24$, $p < 0.01$ Environment: $\eta^2 = 0.01$, $F = 0.03$, $p = 0.97$ BoMa: $\eta^2 = 0.10$, $F = 12.77$, $p < 0.01$ T-E: $\eta^2 = 0.04$, $F = 0.93$, $p = 0.45$ T-BoMa: $\eta^2 = 0.02$, $F = 1.34$, $p = 0.26$
2 (data not transformed)						

522	D = 0.12 p = 0.02	D = 0.91 P = 0.46	Environment: F = 3.05, p = 0.050 Treatment: F = 0.50, p = 0.61	W = 0.90 p < 0.01	Treatment: η^2 = 0.12, F = 6.81, p < 0.01 Environment: η^2 = 0.01, F = 0.03, p = 0.97 BoMa: η^2 = 0.11, F = 12.80, p < 0.01 T-E: η^2 = 0.03, F = 0.59, p = 0.67 T-BoMa: η^2 = 0.01, F = 0.87, p = 0.42

Table S11 (continued)

Model	AIC	KS Test	Dispersion	Levene's Test	Shapiro-Wilk Test	Fixed Effects
3 (log-transformed)						
	193	D = 0.10 p = 0.11	D = 0.87 p = 0.32	Environment: F = 1.75, p = 0.18 Treatment: F = 2.83, p = 0.06	W = 0.98 p = 0.06	Treatment: η^2 = 0.20, F = 13.30, p < 0.01 Environment: η^2 = 0.01, F = 0.04, p = 0.96 BoMa: η^2 = 0.09, F = 11.93, p < 0.01 T-E: η^2 = 0.05, F = 1.30, p = 0.28 T-BoMa: η^2 = 0.04, F = 2.45, p = 0.09
4 (log-transformed)						
	191	D = 0.10 p = 0.16	D = 0.90 p = 0.42	Environment: F = 1.75, p = 0.18 Treatment: F = 1.95, p = 0.15	W = 0.98 p = 0.02	Treatment: η^2 = 0.30, F = 22.75, p < 0.01 Environment: η^2 = 0.01, F = 0.02, p = 0.98 BoMa: η^2 = 0.08, F = 10.09, p < 0.01 T-E: η^2 = 0.02, F = 0.44, p = 0.78

Table S12. Summary of general linear mixed model diagnostics of fixed effects and interactions using a Gaussian distribution of square root transformed body mass-specific metabolic rate as the dependent variable. AIC, a KS Test, model dispersion, a Levene's Test and a Shapiro-Wilks Test were used to evaluate model fit. All models included colony identity as a random effect. Fixed effects and

interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Treatment (T), Colony-collection environment (E, Environment). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interacting effect. The model was simplified by sequentially dropping non-significant terms; each subsequent model is described in a separate row of the table. The simplest model for which assumptions were met is indicated with an asterisk and used to interpret results (Tables 2-3 in the main manuscript).

Model	**AIC**	**KS Test**	**Dispersion**	**Levene's Test**	**Shapiro-Wilk Test**	**Fixed Effects**
1	148	D = 0.06 p = 0.57	D = 0.90 p = 0.46	Environment: F = 1.01 p = 0.37 Treatment: F = 0.16. p = 0.85	W = 0.98 p = 0.10	Treatment: $\eta^2 = 0.29$, F = 17.09, $p < 0.01$ Environment: $\eta^2 = 0.29$, F = 0.93, p = 0.45 T-E: $\eta^2 = 0.03$, F = 0.49, p = 0.74
2*	126	D = 0.05 p = 0.81	D = 0.94 p = 0.61	Environment: F = 0.89, p = 0.41 Treatment: F = 0.29. p = 0.75	W = 0.99 p = 0.14	Treatment: $\eta^2 = 0.34$, F = 19.21, $p < 0.01$ Environment: $\eta^2 = 0.30$, F = 0.96, p = 0.44 No Interactions

Table S13. Summary general linear mixed model diagnostics, fixed effects and interactions using a negative-binomial distribution of worker movement during respirometry as the dependent variable. AIC, KS Test, and model dispersion were used to evaluate model fit. All models included colony identity as a random effect. Fixed effects and interactions included in the model are provided in the right-most column. Abbreviations for fixed effects and interactions, indicated by a hyphen, are as follows: Colony-collection environment (E, Environment). η^2 indicates eta^2 partial effect size. Significant results ($p < 0.05$) indicate a significant main or interaction effect. The model was simplified by sequentially dropping non-significant terms; each subsequent model is described in a separate row of the table. The simplest model for which assumptions were met is indicated with an asterisk (*) and used to interpret results (Tables 2-3 in the main manuscript).

Model	AIC	KS Test	Dispersion	Fixed Effects
1	1579	D = 0.05 p = 0.91	D = 0.91 p = 0.80	Treatment : $\chi^2 = 9.95$, $p < 0.01$ Environment : $\chi^2 = 4.09$, $p = 0.13$ T-E : $\chi^2 = 2.63$, $p = 0.62$
2*	1573	D = 0.05 p = 0.84	D = 0.93 p = 0.82	Treatment : $\chi^2 = 32.57$, $p < 0.01$ Environment : $\chi^2 = 8.76$, $p = 0.01$ No interaction

Table S14: SMA regressions of CO_2 leaks over time for each respirometry chamber. Regression parameters were used to estimate the volume of CO_2 leaked when workers were in sealed chambers to calculate actual worker CO_2 emission rates.

Syringe ID	n	Y-intercept	Slope	r^2	p value
1	12	7.28×10^{-5}	2.25×10^{-6}	0.687	< 0.01
2	13	8.05×10^{-5}	1.78×10^{-6}	0.295	0.06
3	14	1.15×10^{-4}	1.63×10^{-6}	0.144	0.18
4	14	1.00×10^{-4}	1.54×10^{-6}	0.269	0.06
5	14	1.28×10^{-4}	8.23×10^{-7}	0.065	0.38
6	13	1.07×10^{-4}	1.10×10^{-6}	0.135	0.22

Table S15. Comparison of model results excluding or including outliers of OCR standard deviation and body metabolic rate. All models include colony as a random effect and are identical to those presented in Table 2 of the main manuscript and to the models indicated by asterisks in Tables S5-S13. Interaction effects are indicated by a hyphen. Abbreviations for fixed effects are as follows: Treatment (T), Colony-collection environment (E, Environment), Body Mass (BoMa), Body Metabolic Rate (BoMr).

		Outliers Excluded		**Outliers Included**	
Response Variable	**Effect**	**Test statistic**	**p value**	**Test statistic**	**p value**
Brain Mass (mg)	Treatment	$\chi^2 = 0.87$	0.65	$\chi^2 = 0.38$	0.83
	Environment	$\chi^2 = 1.50$	0.47	$\chi^2 = 1.13$	0.57
	Body Mass	$\chi^2 = 13.62$	< 0.01	$\chi^2 = 14.19$	< 0.01
	T – BoMa	$\chi^2 = 7.09$	0.03	$\chi^2 = 7.68$	0.02
Body Mass (mg)	Treatment	F = 30.48	< 0.01	F = 34.61	< 0.01
	Environment	F = 2.14	0.16	F = 2.75	0.11
Relative Brain Mass (%)	Treatment	$\chi^2 = 33.25$	< 0.01	$\chi^2 = 29.91$	< 0.01
	Environment	$\chi^2 = 30.63$	< 0.01	$\chi^2 = 29.30$	< 0.01
Relative Brain Metabolic Rate (%)	Treatment	F = 10.94	< 0.01	F = 13.56	< 0.01
	Environment	F = 0.20	0.82	F = 0.14	0.87
	Body Mass	F = 5.79	0.02	F = 4.48	0.04
	Brain Mass	F = 0.62	0.43	F = 0.01	0.92
	T – E	F = 1.70	0.15	F = 1.94	0.12
	T – BoMa	F = 1.54	0.22	F = 1.57	0.22
Brain Metabolic Rate (μW)	Treatment	F = 1.17	0.31	F = 0.77	0.47
	Environment	F = 0.26	0.77	F = 0.30	0.75
	BoMa	F = 0.52	0.47	F = 0.83	0.36
	T – E	F = 3.42	0.01	F = 3.56	< 0.01
	T – BoMa	F = 4.37	0.01	F = 4.16	0.02
Brain Mass-specific Metabolic Rate ($\mu W\ mg^{-1}$)	Treatment	F = 1.20	0.31	F = 0.94	0.39
	Environment	F = 0.81	0.51	F = 0.66	0.55
	Body Mass	F = 4.56	0.04	F = 4.48	0.04
	T – E	F = 6.31	< 0.01	F = 6.97	< 0.01
	T – BoMa	F = 8.04	< 0.01	F = 8.34	< 0.01
Body Metabolic Rate (μW)	Treatment	F = 10.24	< 0.01	F = 12.79	< 0.01
	Environment	F = 0.03	0.97	F = 1.02	0.44
	Body Mass	F = 12.77	< 0.01	F = 8.30	< 0.01
	T – E	F = 0.93	0.45	F = 1.17	0.33
	T – BoMa	F = 1.34	0.26	F = 1.43	0.24

Table S15 (continued)

		Outliers Excluded		Outliers Included	
Response Variable	**Effect**	**Test statistic**	**p value**	**Test statistic**	**p value**
Body Mass-specific Metabolic Rate (μW mg-1)	Treatment	$F = 19.21$	< 0.01	$\chi^2 = 24.01$	< 0.01
	Environment	$F = 0.96$	0.44	$\chi^2 = 1.91$	0.28
Movement Distance (mm)	Treatment	$\chi^2 = 32.57$	< 0.01	$\chi^2 = 34.67$	< 0.01
	Environment	$\chi^2 = 8.76$	0.01	$\chi^2 = 6.22$	0.045

SI References

1. P. E. Schilman, F. Roces. Foraging energetics of a nectar-feeding ant: metabolic expenditure as a function of food-source profitability. *J. Exp. Biol*. **209**, 4091-4101 (2006).

2. S. Sanada-Morimura, T. Satoh, Y. Obara, Territorial behavior and temperature preference for nesting sites in a pavement ant Tetramorium tsushimae. *Insect. Soc*. **53**, 141-148 (2006).

3. G. Wegener, "Insect brain metabolism under normoxic and hypoxic conditions" in Arthropod Brain: Its evolution, structure, development and function, A. P. Gupta, Ed. (John Wiley & Sons, 1987), pp. 369-397.

4. M. G. Nielsen, T. F. Jensen, I. B. Holm-Jensen, Effect of load carriage on the respiratory metabolism of running worker ants of *Camponotus herculeanus* (Formicidae). *Oikos* **39**, 137-142 (1982).

5. M. G. Nielsen, C. Baroni-Urbani, Energetics and foraging behaviour of the European seed harvesting ant *Messor capitatus*. I. Respiratory metabolism and energy consumption of unloaded and loaded workers during locomotion. *Physiol. Entomol*. **15,** 441-448 (1990).

6. M. G. Nielsen, K.A. Christian, The mangrove ant, *Camponotus anderseni*, switches to anaerobic respiration in response to elevated CO_2 levels. *J. Insect. Physiol.* **53**, 505-508 (2007).

7. F. D. Duncan, J. R. B. Lighton. The burden within: the energy cost of load carriage in the honeypot ant *Myrmecocystus*. *Physiol. Zool*. **67**, 190-203 (1994).

8. J. T. Vogt, A. G. Appel. Standard metabolic rate of the fire ant, *Solenopsis invicta* Buren: effects of temperature, mass, and caste. *J. Insect Physiol.* **45**, 655-666.

9. J. R. B. Lighton, *Measuring Metabolic Rates: A Manual for Scientists* (Oxford University Press, 2019).

10. A. Buschiazzo *et al*., Effect of starvation on brain glucose metabolism and 18F-2-fluoro-2-deoxyglucose uptake: an experimental in-vivo and ex-vivo study. *EJNMMI Res.* **8**, 44 (2018).

11. S. Oikawa, Y. Itazawa, Relationship between metabolic rate in vitro and body mass in a marine teleost, porgy Pagrus major. Fish Physiol. Biochem **10**, 177–182 (1992)

12. S. Oikawa, Y. Itazawa, Relationship between summated tissue respiration and body size in a marine teleost, the porgy Pagrus major. Fisheries Sci **69**, 687-694 (2003).

13. D. I. Warton, R. A. Duursma, D. S. Falster, S. Taskinen, smatr 3 - an R package for estimation and inference about allometric lines. *Methods Ecol. Evol*. **3**, 257-259 (2012).

14. F. Hartig, DHARMa: residual diagnostics for hierarchical (multi-level/mixed) regression modls. *R Package version 020*. (2018)

15. M. S. Ben-Shachar, D. Lüdecke, D. Makowski, effectsize: Estimation of effect size indicies and standardized parameters. *J. Open Source Softw*. **5,** 12815 (2020).

16. M. E. Brooks *et al.,* glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *R. J*. **9**, 378-400 (2017).

17. E. Csata, A. Dussutour, Nutrient regulation in ants (Hymenoptera: Formicidae): a review. *Myrmecol. News* **29**, 111-124 (2019).